

How Long Do Markets Need to Fully React to Monetary Policy Announcements?

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[†]Views expressed are my own and are not necessarily those of the US Treasury or the US government.

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Motivation

- ▶ News is released → Financial markets react to news
 - If change in price $\approx$ change in expectations → Unanticipated news

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- ▶ Useful for empirical macro: Obtain shocks from news to infer causal effects
 - Ex: Monetary policy (MP) announcements

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- ▶ Useful for empirical macro: Obtain shocks from news to infer causal effects
- ▶ Method: High-frequency Identification of MP surprises
 - Measure price change **within event window** around MP announcement
 - Most popular choice in literature: **30 minutes**

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- ▶ Method: High-frequency Identification of MP surprises
- ▶ **Research Q:** What size should the window length around MP announcements be?
 - Too short: Markets might **not fully react** to policy news yet
 - Too long: Change in price $\approx$ MP shocks w/ **unrelated news, confounding factors**
 - **Just right:** Change in price $\approx$ MP shocks with **minimised noise**

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- ▶ **Wrong A:** Contributes to MP shocks lacking precision because of noise

This Paper

Exogeneity/Noise Trade-off

UAT + Layers

- ▶ **This Paper:** Estimate **optimal window size** for FOMC statements by:
 - **Natural language processing (NLP)** to combine price dynamics with **text-based signal**

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 - NLP approximation $\rightarrow$ Text-based signal = $\Delta \widehat{\text{Asset prices}}$ within **given event window**

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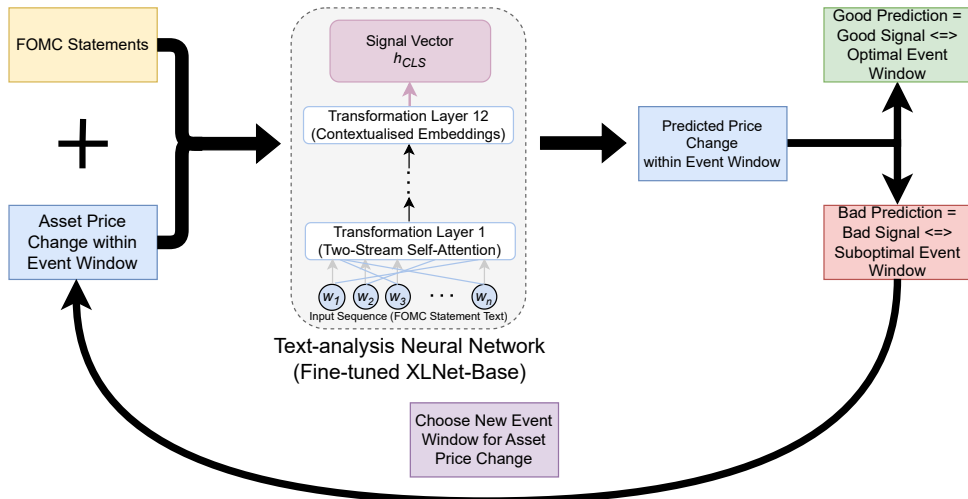
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 - Approximating **underlying** relationship $f(\text{FOMC statement}) = \Delta \text{Asset prices}$
 - NLP approximation $\rightarrow$ Text-based signal = $\widehat{\Delta \text{Asset prices}}$ within **given event window**
- 1. **Optimal window only:** Noise has **min average impact** on $\Delta \text{asset prices}$
- 2. **Optimal window only:** NLP text-based signal has **highest** precision
- 3. **Any other window:** Bad approximation by NLP $\rightarrow$ **Bad signal**

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Exogeneity/Noise Trade-off UAT + Layers

- **This Paper:** Estimate **optimal window size** for FOMC statements using NLP



Previous of Results: Summary

FOMC Statement Ex

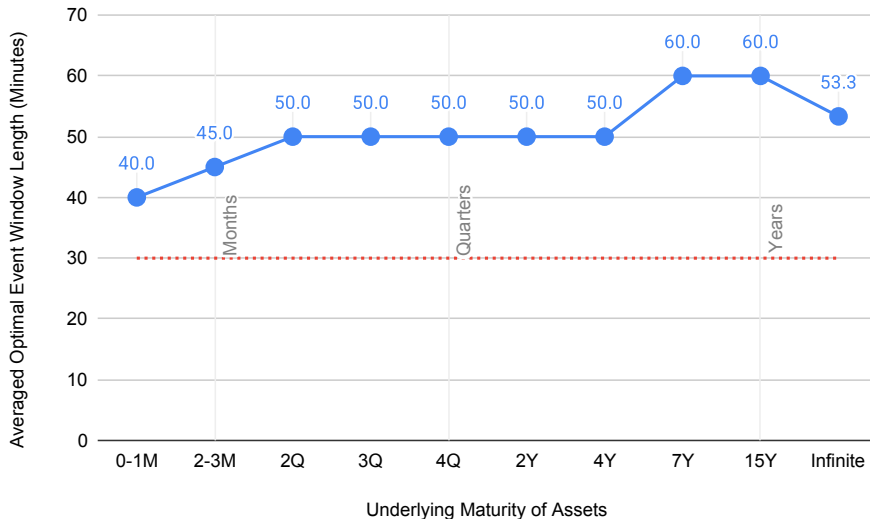
- ▶ **How Long?** Longer than literature standard of 30 minutes:
 - On avg, markets fully react within window 10 min before and 30+ min after
 - Underlying maturity of assets $\uparrow$, then average optimal window length $\uparrow$
 - Underlying maturity of asset at least 2 quarters out $\rightarrow$ 50- to 60-min window
 - Complex/dissimilar/dissent statements $\rightarrow$ Relatively longer windows

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 - Complex/dissimilar/dissent statements $\rightarrow$ Relatively longer windows
- ▶ **Effects on MP:** By changing only event window choice:
 - Underlying maturity of assets $\uparrow$, then correlation $\downarrow$ between MP surprise sets
 - MP shocks about forward guidance have $\uparrow$ impact on yields, inflation, and stock prices
 - MP shocks about forward guidance are $\uparrow$ precise on macroeconomic variables

Preview of Results: Visual

Summary Text Summary Table Liquidity



Related Literature and Contributions

[This Paper](#)[Results Preview](#)[Summary Visual](#)

1. Measuring Appropriate Event Window Lengths

- Hillmer and Yu (1979); Chang and Chen (1989); Krivin et al. (2003); Das and King (2021); An et al. (2025); Casini and McCloskey (2025); Jensen (2025); Boehm and Kroner (2026); and others...
- **Contributions:** (1) If news = text $\rightarrow$ Method can estimate optimal window; (2) Introduce noise min. approach

Related Literature and Contributions

[This Paper](#)[Results Preview](#)[Summary Visual](#)

1. Measuring Appropriate Event Window Lengths

2. Text Analysis in Monetary Policy Communication

- Lucca and Trebbi (2009); Hansen and McMahon (2016); Hansen, McMahon, and Prat (2017); Cieslak and Vissing-Jorgensen (2020); Husted et al. (2020); Handlan (2022a); Handlan (2022b); Acosta (2023); Doh et al. (2023); Cieslak and McMahon (2023); Cieslak, Hansen, et al. (2023); Gorodnichenko et al. (2023); Aruoba and Drechsel (2024); Gáti and Handlan (2025a); Gáti and Handlan (2025b); Piller et al. (2025); and others...
- **Contributions:** ↑ adoption of NLP to quantify dimensions of MP communication

Related Literature and Contributions

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1. Measuring Appropriate Event Window Lengths

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3. Event Window Lengths in Monetary Policy

- Examples: Gürkaynak, Sack, et al. (2005); Gertler and Karadi (2015); Nakamura and Steinsson (2018); Gürkaynak, Kisacikoğlu, et al. (2020); Swanson and Jayawickrema (2023); Handlan (2022b); Bauer and Swanson (2023a); An et al. (2025); Boehm and Kroner (2026); and others...
- **Contributions:** (1) Optimal window length around FOMC statements > 30-min; (2) different markets, different window lengths; (3) MP effects are amplified and more precise

Presentation Roadmap

- ① Introduction
- ② Motivating Framework
- ③ Optimal Event Windows
- ④ MP Surprises & Shocks
- ⑤ Conclusion

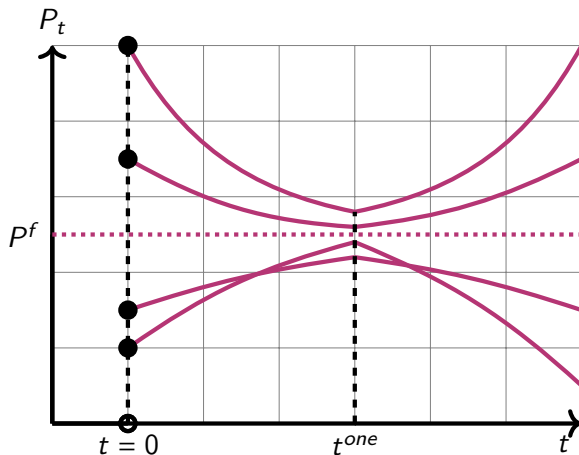
Motivating Framework of Asset Market Prices

$$\{P_t = P_t^f + \varepsilon_t^c + \varepsilon_t^n | t \geq 0\}, \quad (1)$$

► **Fundamental Price: Constant**

$$\rightarrow P_t^f = P^f \in \mathbb{R}$$

- Price **because of** news



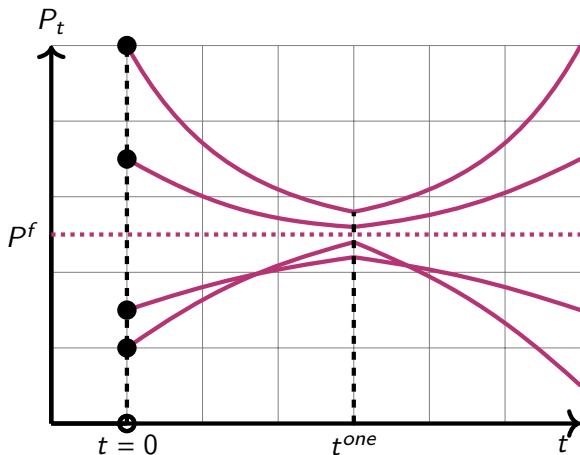
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- ▶ **Cognitive Noise:** Decaying AR(1)
 $\rightarrow \varepsilon_t^c = \rho^c \varepsilon_{t-1}^c + e^{-\mathcal{D}t} v_t^c$

Interpretations

- $v_t^c \sim \mathcal{N}(0, \sigma_c^2)$
- $|\rho_c| < 1$
- Decay: $\mathcal{D} \in \mathbb{R}^+$
- **Assumption:** $\text{Var}(\varepsilon_0^c) = \sigma_c^2$

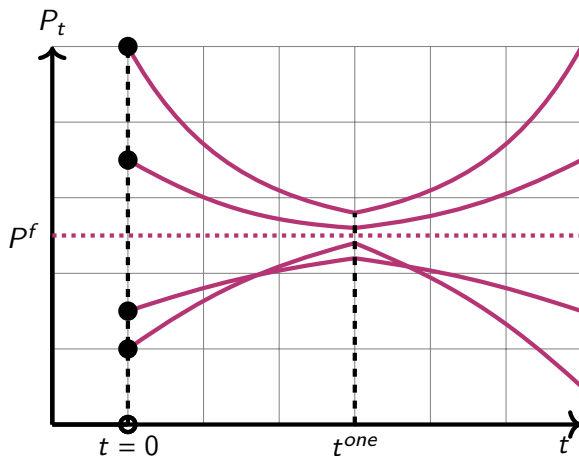


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Interpretations
- ▶ **Unrelated News:** Random Walk
 $\rightarrow \varepsilon_t^n = \varepsilon_{t-1}^n + v_t^n$
 - $v_t^n \sim \mathcal{N}(0, \sigma_n^2)$
 - **Assumption:** $\text{Var}(\varepsilon_0^n) = 0$



Single and Multiple News

Single News Derivations

Multiple News Derivation

▶ **Single News ($N = 1$):**

- Price variance evolves as **cognitive noise** decays and **unrelated news** accumulates
- **Goal:** Find **minimising time t^{one}**

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[†]I numerically verify the dynamics of the t^{one} for various values of σ_c^2 and σ_n^2 in its indirect expression, holding the other parameters constant.

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Signal Visual Ex

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- **Strategy:** Find **MSE-minimising time** t^* relative to signal $\rightarrow s_i$ **precision matters:**

Signal Visual Ex

$$t^* : \min_t \frac{1}{N} \sum_{i=1}^N (P_{i,t} - s_i)^2 \implies \min_t \left[\frac{1}{N} \sum_{i=1}^N (\varepsilon_{i,t}^c + \varepsilon_{i,t}^n)^2 + \sigma_s^2 \right] \quad (2)$$

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Motivating Framework Takeaways

[Simulation Results](#) [Exogeneity/Noise Trade-off](#)

- ▶ **Summary:** Always short, always long window → **Bad idea**
 - “Good” signal → Possible to estimate time horizon reflecting full reactions
- ▶ **Constraint:** MP shocks = **Finite sample** problem → “**Good**” signal matters

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- ▶ **Reality:** **Do not know** noise processes
 - Framework uses strong assumptions to illustrate points
- ▶ **Solution:** NLP method estimates optimal window **without knowing**
 - NLP method can approximate **underlying relationship**

Presentation Roadmap

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Estimating Optimal Event Windows: Method

Exogeneity/Noise Trade-off

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FOMC Statement Text Prep

FOMC Statement Ex

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FOMC Statement Text Prep

FOMC Statement Ex Why FOMC Statements?

Futures List

Approach Input/Output Visual

- **Nonparametric regression** approximated by many linear + non-linear combinations
- Neural network (NN) approximation $\rightarrow$ Text-based signal = Δ Asset prices

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 - **Nonparametric regression** approximated by many linear + non-linear combinations
 - Neural network (NN) approximation $\rightarrow$ Text-based signal = $\Delta\text{Asset prices}$
 - ▶ **Strategy:** Regress Δ asset prices in **different** event windows on FOMC statements
 - Pick event window where NN has $\uparrow$ **avg out-of-sample predictive performance**
 - **Optimal window only:** Noise components have **min average impact** on Δ asset prices
- $\Rightarrow$ “Jointly” estimate optimal window **and** “good” signal

FOMC Statement Text Prep

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Approach Input/Output Visual

CV Strategy CV Visual Method “Loop” Method Visual

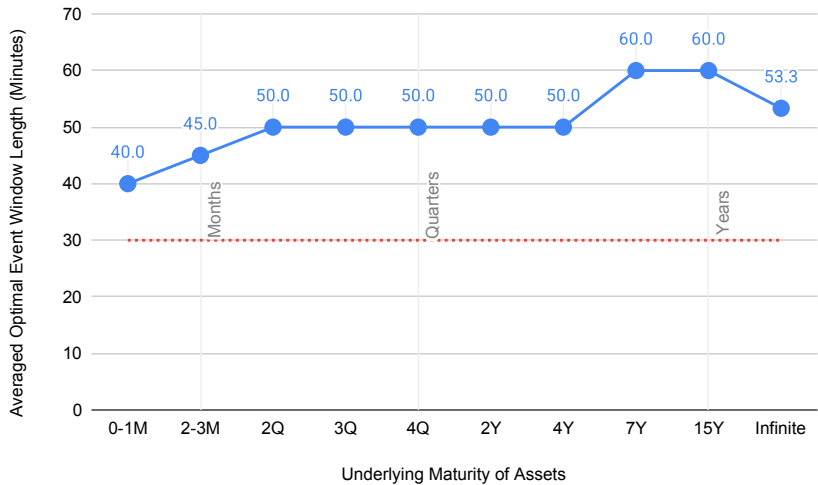
 R^2_{OOS}

Optimal Event Windows: Main Results

Summary Visual (Intro)

FF3 USc2

► How Long?



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[FF3](#) [USc2](#)

- On avg, markets fully react within window 10 min before and **30+ min after**
- Underlying maturity of assets **increases** → Avg optimal window length **increases**
[Other Assets](#)
- Underlying maturity of asset **at least 2 quarters out** → **50–60-min** window

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► How Much Better?

Summary Table

- OOS predictive quality $\approx$ NN generalisability $\uparrow$ by 2–17% when window $\uparrow$ to 40+ min
 R^2_{OOS}
- All event windows → Positive $\overline{R^2_{OOS}}$ → XLNet-Base can approx underlying relationship

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► Why?

[FF3 Scatter](#) [USc2 Scatter](#)

- On avg, markets **under-react**, ex-post, to FOMC statement within standard windows
- “Soft” information = Longer to process → Info asymmetry to resolve [FOMC Statement Ex](#)
 - Post-FOMC drift: Indriawan et al. (2021); Brooks et al. (2023)
 - Fed Info/RTN Effect: Nakamura and Steinsson (2018); Bauer and Swanson (2023b)
 - Investor heterogeneity/Sequential learning: Ai et al. (2021)
 - Macro attention/Risk premia: Fisher et al. (2022)

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Monetary Policy Surprises: Correlations Along the Yield Curve

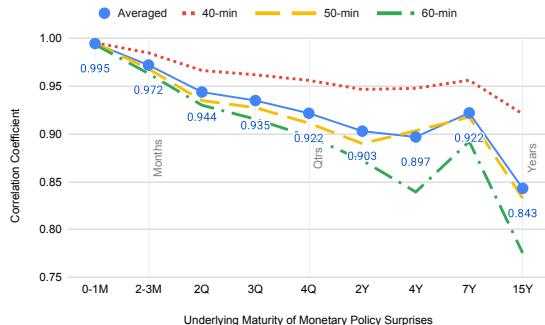
Construction Details

► Methodology:

1. Pick optimal window: 40-, 50-, 60-min
2. Construct MP surprises within **30-min** and **optimal** windows DP → MP Surprise
 - Using FFFs: $mp1, mp2$
 - Using Eurodollar futures: $\Delta ed2, \Delta ed3, \Delta ed4$
 - Using Treasury futures: $\Delta t2, \Delta t5, \Delta t10, \Delta t30$
3. Calculate correlations between MP surprise sets

► Result (Figure):

- Changing only window length has $\uparrow$ effect at **farther horizons**



Monetary Policy Shocks: Construction Methods

- ▶ Focus on **median** optimal event window length: **50 minutes**

Monetary Policy Shocks: Construction Methods

- ▶ Focus on **median** optimal event window length: **50 minutes**
- ▶ Use **full set** of MP surprises as instruments
 - Prevent dampening of MP during ELB period (Brennan et al., 2024; An et al., 2025)

Monetary Policy Shocks: Construction Methods

- ▶ Focus on **median** optimal event window length: **50 minutes**
- ▶ Use **full set** of MP surprises as instruments
- ▶ Construct MP shocks using diff methods within 30-minutes and optimal windows:

PCA MP Shock Visuals Summary Stats

1. Gürkaynak, Sack, et al. (2005):

- $GSS_T \rightarrow 1^{st}$ Principal component rotated to **drive** $mp1$
- $GSS_P \rightarrow 2^{nd}$ Principal component rotated to have **no effect** on $mp1$

2. Nakamura and Steinsson (2018):

- $NS_{MP} \rightarrow 1^{st}$ Principal component of MP surprises

3. Jarociński and Karadi (2020):

- $JK_{MP} \rightarrow 1^{st}$ Principal component of MP surprises w/ SPX + co-movement
- $JK_{CBI} \rightarrow 1^{st}$ Principal component of MP surprises w/ SPX + co-movement

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- ▶ **All shocks scaled:** 1bp $\uparrow$ in shock $\rightarrow$ 1bp $\uparrow$ in nominal 1-year Treasury yield

MP Shocks: Effects on Interest Rates, Break-even Inflation, Equities (1/2)

► LHS Variables:

1. ΔTY^i = Daily change in nominal Treasury yields, $i \in \{1, 2, 5, 10\}$
2. $\Delta TIPS^i$ = Daily change in Treasury Inflation-Protected Security yields, $i \in \{2, 5, 10\}$
3. ΔBEI^i = Daily change in end-of-day break-even inflation, $i \in \{2, 5, 10\}$
4. DP_{SPX} = Price log-difference of *SPX* within 30-minute and optimal windows
5. DP_{ESc1} = Price log-difference of *ESc1* within 30-minute and optimal windows
6. DP_{ESc2} = Price log-difference of *ESc2* within 30-minute and optimal windows

► Specification:

$$y^j = \beta_0^{j,k,l} + \beta_1^{j,k,l} (Shock)^{k,l} + \varepsilon^{j,k,l}, \quad (3)$$

- Shock $k \in \{GSS_T, GSS_P, NS_{MP}, JK_{MP}, JK_{CBI}\}$
- Event window $l \in \{30 \text{ minutes, optimal}\}$

MP Shocks: Effects on Interest Rates, Break-even Inflation, Equities (2/2)

- ▶ **Using Optimal Windows:** MP shocks about forward guidance have **↑ effects** on:
 - Nominal interest rates by **8bp on avg** [Reg Table](#)
 - Real interest rates by **14bp on avg** [Reg Table](#)
 - Break-even inflation by **-5bp on avg** [Reg Table](#)
 - Stock prices and E-mini futures by **14% on avg** [Reg Table](#)
- ▶ Using non-optimal windows → **Attenuated** MP shock effects on financial variables

MP Shocks: Impulse Responses from Local Projection Approach (1/2)

- ▶ **LHS Variables:** Monthly variables i from FRED [Summary Stats](#)
 - Log Consumer Price Index (CPI)
 - Log Industrial Production (IP)
 - Nominal 2-year Treasury yield
 - Excess Bond Premium (EBP) (Gilchrist and Zakrajšek, 2012)
- ▶ **Specification:** Lag-augmented local projections (Olea and Plagborg-Møller, 2021)

$$y_{t+h}^{i,l} = \theta^{i,k,l} (Shock)_t^{k,l} + controls + \eta^{i,k,l} \quad (4)$$

- ▶ Shock $k \in \{GSS_T, GSS_P, NS_{MP}, JK_{MP}, JK_{CBI}\}$
 - All shocks converted to monthly frequency (No meeting $\rightarrow$ Zero shock) [Summary Stats](#)
- ▶ Event window $l \in \{30 \text{ minutes, optimal}\}$
- ▶ **All shocks scaled:** 100bp $\uparrow$ shock $\rightarrow$ Contractionary shock

MP Shocks: Impulse Responses from Local Projection Approach (2/2)

- ▶ Calculate **confidence interval (CI) widths** under 30-minute and optimal windows
 - Average ratio of CI widths: **0.9173** → Average CI width ↓ by **8%**
 - Median ratio of CI widths: 0.9410
- ▶ **Using Optimal Windows:** Impulse responses of macro variables have ↑ **precision**
LP Visuals
- ▶ Using non-optimal windows: Increased **noise** → MP shocks have ↓ **relevancy**

Extensions: How do Statement Characteristics Affect Window Length?

- ▶ **Question:** Do FOMC statement **characteristics** affect **optimal** window length?
- ▶ **Approach:** Condition estimation on specific statement features under “One Signal”
One Signal Approach
- ▶ **Findings:** On average, **longer** windows associated with: FOMC Statement Ex Characteristics
 - More **semantically complex** statements (Reading Level $\uparrow$) Complexity
 - Statements **less similar** to previous statement (Cosine Similarity $\downarrow$) Similarity
 - Statements with presence of **dissents** in the voting record Dissents

Presentation Roadmap

- ① Introduction
- ② Motivating Framework
- ③ Optimal Event Windows
- ④ MP Surprises & Shocks
- ⑤ Conclusion

Conclusion

- ▶ **This Paper:** Estimate **optimal window size** for FOMC statements by:
 - Combining **text-analysis neural network** with observed financial price dynamics
 - Approximating **underlying** relationship $f(\text{FOMC statement}) = \Delta \text{Asset prices}$

Conclusion

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 - Approximating **underlying** relationship $f(\text{FOMC statement}) = \Delta \text{Asset prices}$
- ▶ **How Long?** **Longer** than literature standard of 30 minutes:
 - On avg, markets fully react within window 10 min before and **30+ min after**
 - Underlying maturity of assets $\uparrow$, then average optimal window length $\uparrow$
 - Underlying maturity of asset at least 2 quarters out $\rightarrow$ **50- to 60-min window**
 - Complex/dissimilar/dissent statements $\rightarrow$ Relatively **longer** windows

Conclusion

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 - Complex/dissimilar/dissent statements $\rightarrow$ Relatively **longer** windows
- ▶ **Effects on MP:** By changing only event window choice:
 - Underlying maturity of assets $\uparrow$, then **correlation** $\downarrow$ between MP surprise sets
 - Correlation can $\downarrow$ as much as **10% for long-maturity assets**
 - MP shocks about forward guidance have **$\uparrow$ impact** on yields, inflation, and stock prices
 - Ex: Average MP shock effect on real interest rates $\uparrow$ by **14bp**
 - MP shocks about forward guidance are **$\uparrow$ precise** on macroeconomic variables
 - Ex: Average impulse response CI width $\downarrow$ by **8%**

Next Steps

► Next steps:

1. Estimate optimal event window lengths for other MP communication
 - Ex: Fed Chair and Vice-chair **speeches** (Swanson and Jayawickrema, 2023)
2. Analyse how deeper changes in MP communication affect optimal windows
 - **Conceptual** complexity effect > semantic complexity effect?
3. Different optimal window lengths for **different states**?
 - Ex: High/low inflationary periods

Thank you!

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Relative Exogeneity and Trade-off with Noise

[This Paper](#)[Framework Takeaways](#)[Method](#)

- ▶ **Problem:** Longer windows can:
 - Introduce unrelated news, confounding factors → Endogeneity bias
- ⇒ Causes issue with identifying MP surprises

Relative Exogeneity and Trade-off with Noise

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 - MP announcement “dominates” all other background noise sources at all time horizons

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- ▶ **Solution:** Relative exogeneity (Casini and McCloskey, 2025)
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- ▶ **Plausible?** Prior studies show this condition remains credible for long windows
 - Ex: Fleming and Piazzesi (2005); Jensen (2025); Boehm and Kroner (2026); and others...

Relative Exogeneity and Trade-off with Noise

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- Ex: Fleming and Piazzesi (2005); Jensen (2025); Boehm and Kroner (2026); and others...

⇒ **This Paper:** Estimating optimal window size for FOMC statements using NLP:

- Minimises impact of noise without introducing significant endogeneity bias

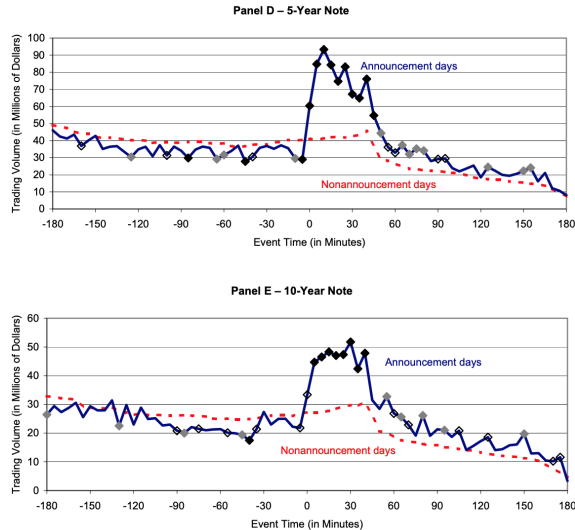
Liquidity: Related Symptom for Longer Event Windows (1/2)

[Summary Visual \(Intro\)](#)

- ▶ Do not currently have data access **BUT**:
 - Fleming and Piazzesi, 2005: $\uparrow$ asset horizon, then $\uparrow$ time length of abn trading volume
 - Kroner, 2025: Within asset types, futures maturity $\uparrow$, then $\downarrow$ change in trading volume
 - ▶ **Both papers**: Document $\uparrow$ trading volume on macro news for longer times
- Assets with $\uparrow$ underlying maturities might need **more time** to **fully react**

Liquidity: Related Symptom for Longer Event Windows (2/2)

Summary Visual (Intro)



► Fleming and Piazzesi, 2005

Cognitive Noise Interpretations

[Back to Framework](#)[Back to Framework Components](#)

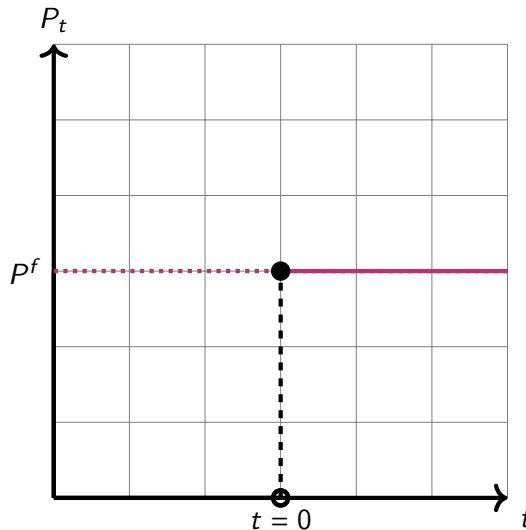
1. **Over-reaction**: De Bondt and Thaler (1985); La Porta (1996); Antweiler and Frank (2022); Da et al. (2011); Bordalo, Gennaioli, La Porta, et al. (2019); Bordalo, Gennaioli, Ma, et al. (2020); Boguth et al. (2023); Bianchi et al. (2024)
2. **Under-reaction**: Bernard and Thomas (1989); LKC Chan et al. (1996); WS Chan (2003); Lucca and Moench (2015); Boguth et al. (2023); Bianchi et al. (2024)
3. **Noise trading**: Bloomfield et al. (2009); Hervé et al. (2019); Ben-David et al. (2022)
4. **Algorithmic trading**: Caivano (2015); Weller (2017); Gider et al. (2019); Bazzana and Collini (2020); Ben Ammar and Hellara (2022)

Visualising How Noise Components Affect Prices (1/3)

[Back to Framework](#)

No cognitive noise + No unrelated news

- ▶ $P_t \rightarrow P^f$ because of no cognitive noise
- ▶ P_t moves anywhere over time because of unrelated news
- Choose shortest event window



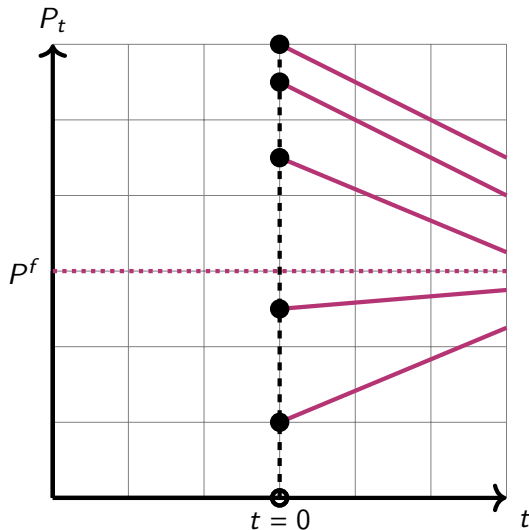
Visualising How Noise Components Affect Prices (2/3)

[Back to Framework](#)

Cognitive noise + No unrelated news

[Interpretations](#)

- ▶ P_t jumps anywhere because of cognitive noise
- ▶ $P_t \rightarrow P^f$ because of no unrelated news
- Choose long event window

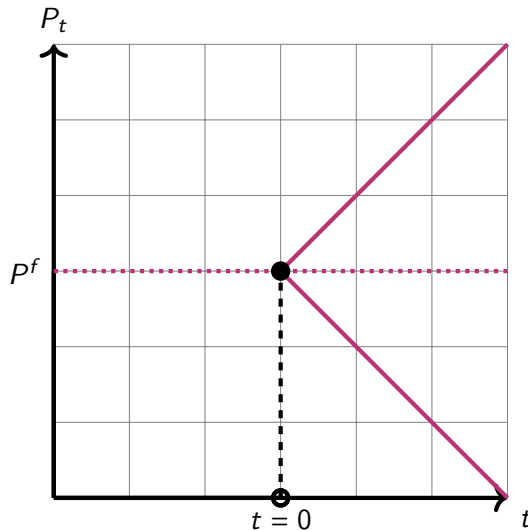


Visualising How Noise Components Affect Prices (3/3)

[Back to Framework](#)

No cognitive noise + Unrelated news

- ▶ $P_t \rightarrow P^f \because$ no cognitive noise
- ▶ P_t moves anywhere over time $\because$ unrelated news
- ▶ $\therefore$ Choose short event window



Derivation of $\text{Var}(P_t|t \geq 0)$ and $\frac{\partial \text{Var}(P_t|t \geq 0)}{\partial t}$ (1/3)

[Back](#)

$$\begin{aligned}\text{Var}(P_0) &= \text{Var}(\varepsilon_0^c) + \text{Var}(\varepsilon_0^n) \\ &= \sigma_c^2\end{aligned}$$

$$\begin{aligned}\text{Var}(P_1) &= \text{Var}(\varepsilon_1^c) + \text{Var}(\varepsilon_1^n) \\ &= \sigma_c^2(\rho_c^2 + e^{-2\mathcal{D}}) + \sigma_n^2\end{aligned}$$

$$\begin{aligned}\text{Var}(P_2) &= \text{Var}(\varepsilon_2^c) + \text{Var}(\varepsilon_2^n) \\ &= \sigma_c^2(\rho_c^4 + \rho_c^2 e^{-2\mathcal{D}} + e^{-4\mathcal{D}}) + 2\sigma_n^2\end{aligned}$$

$$\vdots$$

$$\text{Var}(P_t|t \geq 0) = \left[\sum_{i=0}^t \rho_c^{2(t-i)} e^{-2\mathcal{D}i} \right] \sigma_c^2 + t\sigma_n^2$$

$$\Rightarrow \text{Var}(P_t|t \geq 0) = \left[\frac{\rho_c^{2(t+1)} - e^{-2(t+1)\mathcal{D}}}{\rho_c^2 - e^{-2\mathcal{D}}} \right] \sigma_c^2 + t\sigma_n^2$$

Derivation of $\text{Var}(P_t|t \geq 0)$ and $\frac{\partial \text{Var}(P_t|t \geq 0)}{\partial t}$ (2/3)

[Back](#)

$$\text{Var}(P_t|t \geq 0) = \left[\frac{\rho_c^{2(t+1)} - e^{-2(t+1)\mathcal{D}}}{\rho_c^2 - e^{-2\mathcal{D}}} \right] \sigma_c^2 + t\sigma_n^2$$

$$\frac{\partial \text{Var}(P_t|t \geq 0)}{\partial t} = \left\{ \frac{2 \left[\ln(\rho_c) \rho_c^{2(t+1)} + \mathcal{D} \left[(e^{-2(t+1)\mathcal{D}}) \right] \right]}{\rho_c^2 - e^{-2\mathcal{D}}} \right\} \sigma_c^2 + \sigma_n^2$$

Derivation of $\text{Var}(P_t|t \geq 0)$ and $\frac{\partial \text{Var}(P_t|t \geq 0)}{\partial t}$ (3/3)

[Back](#)

$$\text{Var}(P_t|t \geq 0) = \overbrace{\left[\frac{\rho_c^{2(t+1)} - e^{-2(t+1)\mathcal{D}}}{\rho_c^2 - e^{-2\mathcal{D}}} \right]}^{\text{lim}_{t \rightarrow \infty} \text{ is } 0} \sigma_c^2 + t\sigma_n^2 \quad (5)$$

§An interior solution (i.e., $t^{one} > 0$) exists if $\frac{\sigma_n^2}{\sigma_c^2} < \frac{2[\ln(\rho_c)\rho_c^2 + \mathcal{D}e^{-2\mathcal{D}}]}{e^{-2\mathcal{D}} - \rho_c^2}$. Intuitively, if background noise accumulates too quickly relative to initial cognitive confusion, this condition is violated, and the optimal horizon collapses to the boundary solution $t = 0$.

Derivation of $\text{Var}(P_t|t \geq 0)$ and $\frac{\partial \text{Var}(P_t|t \geq 0)}{\partial t}$ (3/3)

[Back](#)

$$\text{Var}(P_t|t \geq 0) = \overbrace{\left[\frac{\rho_c^{2(t+1)} - e^{-2(t+1)\mathcal{D}}}{\rho_c^2 - e^{-2\mathcal{D}}} \right]}^{\text{lim}_{t \rightarrow \infty} \text{ is } 0} \sigma_c^2 + t\sigma_n^2 \quad (5)$$

$$\Rightarrow t^{\text{one}} : \mathcal{D} [e^{-2(t+1)\mathcal{D}}] + \ln(\rho_c) \rho_c^{2(t+1)} = \left[\frac{(e^{-2\mathcal{D}} - \rho_c^2)}{2} \right] \frac{\sigma_n^2}{\sigma_c^2} \S \quad (6)$$

$\S$ An interior solution (i.e., $t^{\text{one}} > 0$) exists if $\frac{\sigma_n^2}{\sigma_c^2} < \frac{2[\ln(\rho_c)\rho_c^2 + \mathcal{D}e^{-2\mathcal{D}}]}{e^{-2\mathcal{D}} - \rho_c^2}$. Intuitively, if background noise accumulates too quickly relative to initial cognitive confusion, this condition is violated, and the optimal horizon collapses to the boundary solution $t = 0$.

Derivation of $\text{Var}(P_t|t \geq 0)$ and $\frac{\partial \text{Var}(P_t|t \geq 0)}{\partial t}$ (3/3)

$$\text{Var}(P_t|t \geq 0) = \overbrace{\left[\frac{\rho_c^{2(t+1)} - e^{-2(t+1)\mathcal{D}}}{\rho_c^2 - e^{-2\mathcal{D}}} \right]}^{\lim_{t \rightarrow \infty} \text{ is } 0} \sigma_c^2 + t\sigma_n^2 \quad (5)$$

$$\Rightarrow t^{one} : \mathcal{D} [e^{-2(t+1)\mathcal{D}}] + \ln(\rho_c) \rho_c^{2(t+1)} = \left[\frac{(e^{-2\mathcal{D}} - \rho_c^2)}{2} \right] \frac{\sigma_n^2}{\sigma_c^2} \S \quad (6)$$

$$\Rightarrow \frac{\partial t^{one}}{\partial \sigma_n^2} < 0, \frac{\partial t^{one}}{\partial \sigma_c^2} > 0 \P$$

$\S$ An interior solution (i.e., $t^{one} > 0$) exists if $\frac{\sigma_n^2}{\sigma_c^2} < \frac{2[\ln(\rho_c)\rho_c^2 + \mathcal{D}e^{-2\mathcal{D}}]}{e^{-2\mathcal{D}} - \rho_c^2}$. Intuitively, if background noise accumulates too quickly relative to initial cognitive confusion, this condition is violated, and the optimal horizon collapses to the boundary solution $t = 0$.

$\P$ Because Equation 3 is unable to have t isolated on one side, I numerically verify the dynamics of the t^{one} for various values of σ_c^2 and σ_n^2 in the indirect expression while holding the other parameters constant.

Derivation of MSE Minimisation Problem with Signal

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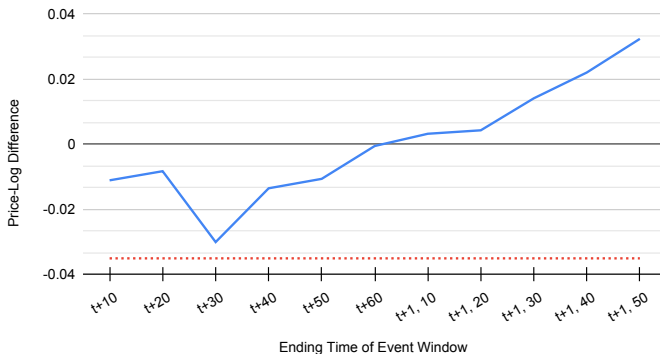
$$\begin{aligned}
 t^* : \min_t \frac{1}{N} \sum_{i=1}^N (P_{i,t} - s_i)^2 &= \min_t \frac{1}{N} \sum_{i=1}^N \left(P_i^f + \varepsilon_{i,t}^c + \varepsilon_{i,t}^n - P_i^f - \xi_i \right)^2 \\
 &= \min_t \frac{1}{N} \sum_{i=1}^N \left(\varepsilon_{i,t}^c + \varepsilon_{i,t}^n - \xi_i \right)^2 \\
 &= \min_t \frac{1}{N} \sum_{i=1}^N \left[\left(\varepsilon_{i,t}^c + \varepsilon_{i,t}^n \right)^2 + \xi_i^2 - 2\xi_i \left(\varepsilon_{i,t}^c + \varepsilon_{i,t}^n \right) \right] \\
 &= \min_t \left\{ \mathbb{E} \left[\left(\varepsilon_{i,t}^c + \varepsilon_{i,t}^n \right)^2 \right] + \mathbb{E} [\xi_i^2] - 2 \mathbb{E} [\xi_i] \mathbb{E} \left[\left(\varepsilon_{i,t}^c + \varepsilon_{i,t}^n \right) \right] \right\} \\
 \implies t^* : \min_t \frac{1}{N} \sum_{i=1}^N (P_{i,t} - s_i)^2 &= \min_t \left[\frac{1}{N} \sum_{i=1}^N \left(\varepsilon_{i,t}^c + \varepsilon_{i,t}^n \right)^2 + \sigma_s^2 \right] \tag{7}
 \end{aligned}$$

Multiple News: Example of Signal in Financial Prices

[Back](#)

- ▶ **Red Dotted Line:** Signal (s_i) $\approx$ Fundamental price change **because of** news
- ▶ **Blue Line:** Observed price change with varying noise levels
- ▶ **Optimisation:** t^* is where signal **best represents** observed price change

Market Price Reactions for S&P 500 Index, 09/08/2011



Simulation Setup (1/3): Initial Conditions

[Back](#)

► $t = 0$: Release of **one** FOMC announcement

- $P_{t,i}^f = P_i^f \in [-100, 100]$
- $\varepsilon_{i,0}^c \in [-100, 100]$
- $\varepsilon_{i,0}^n = 0$
- $\sigma_s \in \mathbb{R}$

Simulation Setup (2/3): MSEs

[Back](#)

- ▶ For single news $i \in N = 10,000$:
 - Simulate $P_{i,t}$ (and components) and s_i up to $t = 100$
 - Calculate $\left(P_{i,t} - P_{i,t}^f\right)^2$ and $(P_{i,t} - s_i)^2$
- ▶ Across all N news:
 - Calculate MSEs $\sum_{i=1}^N \frac{1}{N} \left(P_{i,t} - P_{i,t}^f\right)^2$ and $\sum_{i=1}^N \frac{1}{N} (P_{i,t} - s_{i,t})^2$
 - Calculate t^* and $\hat{t}$

Simulation Setup (3/3): Market Scenarios

[Back](#)

- Calculate $t^*, \hat{t}$ under 3 asset market scenarios:

	Scenario 1	Scenario 2	Scenario 3
P_i^f	$\in [-100, 100]$	$\in [-100, 100]$	$\in [-100, 100]$
$\varepsilon_{i,0}^c$	$\in [-100, 100]$	$\in [-100, 100]$	$\in [-100, 100]$
$\varepsilon_{i,0}^n$	0	0	0
σ_c	100	1	50
$\mathcal{D}$	0.5	2	0.75
σ_n	0.1	1.13	1
ρ_c	0.47	0.47	0.47
σ_s	$\in \mathbb{R}$	$\in \mathbb{R}$	$\in \mathbb{R}$

Table 1: Framework Parameters for Simulations

- Scenario 1 ~ High presence of cognitive noise, little unrelated news
- Scenario 2 ~ Little cognitive noise, high presence of unrelated news
- Scenario 3 ~ Presence of both cognitive noise and unrelated news

Simulation Results

[Back](#)

	Scenario 1	Scenario 2	Scenario 3
<i>Simulation Parameters</i>			
P_i^f	$\in [-100, 100]$	$\in [-100, 100]$	$\in [-100, 100]$
$\varepsilon_{i,0}^c$	$\in [-100, 100]$	$\in [-100, 100]$	$\in [-100, 100]$
$\varepsilon_{i,0}^n$	0	0	0
σ_c	100	1	50
$\mathcal{D}$	0.5	2	0.75
σ_n	0.1	1.13	1
ρ_c	0.47	0.47	0.47
σ_s	$\in \mathbb{R}$	$\in \mathbb{R}$	$\in \mathbb{R}$
<i>Simulation Results</i>			
t^*	16	5	10
$\hat{t}$	15	5	10

Table 2: Framework Parameters and Results from 10,000 Simulations

Preprocessing FOMC Statement Text

[Back to Method](#)

▶ Remove:

- URLs and hyperlinks from statement's HTML file
- FOMC member voting record from end of statement
- List of regional bank request approvals
- Release timestamp (e.g., "For immediate release")

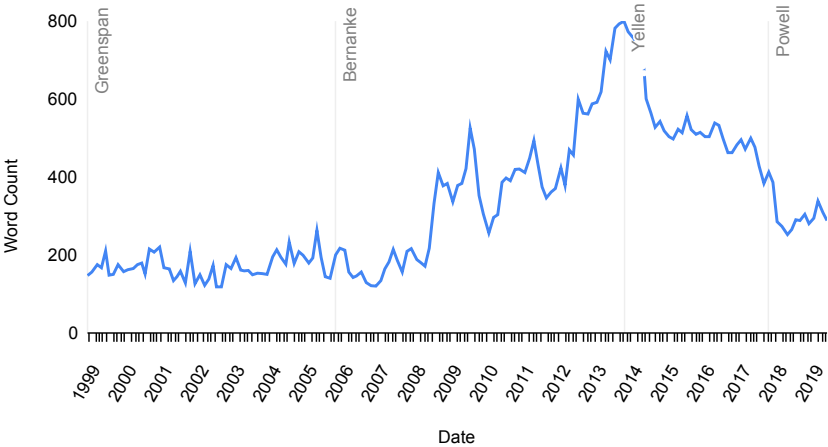
▶ Change:

- Statement file type to text
- Text coding into standardised UTF-8 format (e.g., change length of "-")
- Spacing between words to be one space

Preprocessed FOMC Statement Length

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Number of Words in FOMC Statements



Cleaned FOMC Statement (08/2011) (1/3)

[Results Preview](#)[Method](#)[Summary Text](#)[Extensions](#)

1. Information received since the Federal Open Market Committee met in June indicates that economic growth so far this year has been considerably slower than the Committee had expected.
2. Indicators suggest a deterioration in overall labor market conditions in recent months, and the unemployment rate has moved up.
3. Household spending has flattened out, investment in nonresidential structures is still weak, and the housing sector remains depressed.
4. However, business investment in equipment and software continues to expand.
5. Temporary factors, including the damping effect of higher food and energy prices on consumer purchasing power and spending as well as supply chain disruptions associated with the tragic events in Japan, appear to account for only some of the recent weakness in economic activity.
6. Inflation picked up earlier in the year, mainly reflecting higher prices for some commodities and imported goods, as well as the supply chain disruptions.

Cleaned FOMC Statement (08/2011) (2/3)

Results Preview

Method

Summary Text

Extensions

- 1.
2. More recently, inflation has moderated as prices of energy and some commodities have declined from their earlier peaks. Longer-term inflation expectations have remained stable.
3. Consistent with its statutory mandate, the Committee seeks to foster maximum employment and price stability.
4. The Committee now expects a somewhat slower pace of recovery over coming quarters than it did at the time of the previous meeting and anticipates that the unemployment rate will decline only gradually toward levels that the Committee judges to be consistent with its dual mandate.
5. Moreover, downside risks to the economic outlook have increased.
6. The Committee also anticipates that inflation will settle, over coming quarters, at levels at or below those consistent with the Committee's dual mandate as the effects of past energy and other commodity price increases dissipate further.
7. However, the Committee will continue to pay close attention to the evolution of inflation and inflation expectations.

Cleaned FOMC Statement (08/2011) (3/3)

Results Preview Method Summary Text Extensions

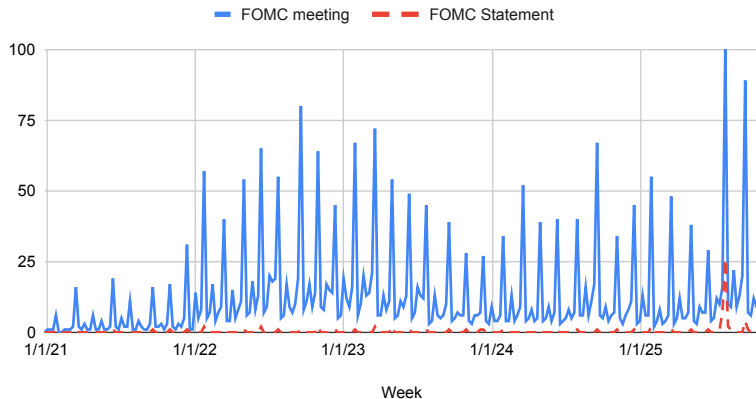
1. To promote the ongoing economic recovery and to help ensure that inflation, over time, is at levels consistent with its mandate, the Committee decided today to keep the target range for the federal funds rate at 0 to 1/4 percent.
2. The Committee currently anticipates that economic conditions—including low rates of resource utilization and a subdued outlook for inflation over the medium run—are likely to warrant exceptionally low levels for the federal funds rate at least through mid-2013.
3. The Committee also will maintain its existing policy of reinvesting principal payments from its securities holdings.
4. The Committee will regularly review the size and composition of its securities holdings and is prepared to adjust those holdings as appropriate.
5. The Committee discussed the range of policy tools available to promote a stronger economic recovery in a context of price stability.
6. It will continue to assess the economic outlook in light of incoming information and is prepared to employ these tools as appropriate.

Why FOMC Statements?

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- ▶ FOMC statements = Initial + **Primary** communication of MP
 - FOMC statement website = 1st – 3rd query on search engines

Google Trends for FOMC Meeting Terms



Output: Interest-rate and Equity Futures

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► **Output:** $DP_{t+n} = \ln\left(\frac{P_{t+n}}{P_{t-10}}\right)$ for interest-rate and equity futures

[Futures Overview](#)

- Price levels at 10-min-intervals: 10 min before to 18 hours after statement release
- Fed Fund Futures: $FF1, FF2, FF3, FF4$
- Eurodollar Futures: $EDcm2, EDcm3, EDcm4$
- 2-Year Treasury Futures: $TUc1, TUc2$
- 5-Year Treasury Futures: $FVc1, FVc2$
- 10-Year Treasury Futures: $TYc1, TYc2$
- 30-year Treasury Futures: $USc1, USc2$
- S&P 500 Index and E-mini Futures: $SPX, ESc1, ESc2$

Futures Contract Overview (1/2)

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- ▶ Federal Fund futures: For a given expiry month, on last day of expiry month, contract pays out 100 minus the average federal funds rate over the final month
- ▶ Eurodollar futures: Quarterly contracts which pay out 100 minus the 3-month US dollar British Banks Association London Interbank Offered Rate (BBA LIBOR) interest rate at the time of expiry month (i.e., March, June, September, and December).
- ▶ 2-year Treasury futures: Quarterly contracts that obligate seller to deliver Treasury notes within a range of maturities up to 2 years on the last day of the expiry month to the buyer (i.e., March, June, September, and December). Average maturity is 2 years (Gürkaynak, Kisacikoğlu, et al., 2020)
- ▶ 5-year Treasury futures: Quarterly contracts that obligate seller to deliver Treasury notes within a range of maturities up to 5 years on the last day of the expiry month to the buyer (i.e., March, June, September, and December). Average maturity is 4 years (Gürkaynak, Kisacikoğlu, et al., 2020)

Futures Contract Overview (2/2)

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- ▶ 10-year Treasury futures: Quarterly contracts that obligate seller to deliver Treasury notes within a range of maturities up to 10 years on the last day of the expiry month to the buyer (i.e., March, June, September, and December). Average maturity is 7 years (Gürkaynak, Kisacikoğlu, et al., 2020)
- ▶ 30-year Treasury futures: Quarterly contracts that obligate seller to deliver Treasury notes within a range of maturities up to 30 years on the last day of the expiry month to the buyer (i.e., March, June, September, and December). Average maturity is 15 years (Gürkaynak, Kisacikoğlu, et al., 2020)
- ▶ S&P 500 E-mini futures: Quarterly contracts that pay out $50 \text{ USD} \times \text{S\&P 500}$ value on the last day of the expiry month (i.e., March, June, September, and December)

NN Input/Output Visual

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- ▶ Each FOMC statement is paired with DP_{t+n} for each asset
- ▶ Input $X_i = 768 \times j$ matrix: Columns = j words in order, rows = 768 word-features

Statement Text

Dec 12, 2006: "The Federal Open Market Committee decided today to keep its target for the federal funds rate at 5 1/4 percent..."

Input Matrix

768 word-features
× 512 words

x_i^1	x_i^2	x_i^3	x_i^4	x_i^5	x_i^6	...	x_i^{512}
The	Federal	Open	Market	Committee	decided		.

- ▶ x_t^0 = Dummy vector that gets updated with intermediate layers of X
- ▶ Output = $DP_{i,t+n}$ for each asset
- ▶ Update XLNet-Base parameters to minimise $\sum_{i \in N} \frac{1}{N} \left(DP_{i,t+n} - \widehat{DP}_{i,t+n} \right)^2$

Popular Text Analysis Methods in Macro

[Back to Approach](#)

1. Counts of single words

- “employment went up, but inflation did not”
- “inflation went up, but employment did not”
- **Problem:** Method produces same measure from both sentences

2. Counts of n-grams

- “economic growth slowed, but is expected to pick up pace later this year”
- **Problem:** Method doesn't quantify full sentence context

Estimating Optimal Event Windows: Approach

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- ▶ **At the Core:** $f(\text{FOMC statement text}) = DP_{t+n}$: **Nonparametric** mapping

Estimating Optimal Event Windows: Approach

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- ▶ **At the Core:** $f(\text{FOMC statement text}) = DP_{t+n}$: **Nonparametric** mapping
- ▶ Popular text analysis methods in empirical macro:
 - “Fitting predictive models on **simple counts of text features**” (Gentzkow et al., 2019)

Estimating Optimal Event Windows: Approach

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- ▶ **At the Core:** $f(\text{FOMC statement text}) = DP_{t+n}$: **Nonparametric** mapping
 - ▶ Popular text analysis methods in empirical macro:
 - “Fitting predictive models on **simple counts of text features**” (Gentzkow et al., 2019)
 - ▶ Popular methods **cannot realistically**:

Popular Method Ex Issues

 - Quantify all info. in text (e.g., Word interdependencies, context, long-term memory)
- ⇒ Approximate $f(\text{FOMC statement text}) = DP_{t+n}$

Estimating Optimal Event Windows: Approach

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- ▶ **At the Core:** $f(\text{FOMC statement text}) = DP_{t+n}$: **Nonparametric** mapping
 - ▶ Popular methods **cannot** quantify $f(\text{FOMC statement text}) = DP_{t+n}$
 - ▶ **Foundation:** Text-analysis neural network **XLNet-Base** (Yang et al., 2019) can:
 - Transfer learning: Fine-tune **pre-trained** XLNet-Base on FOMC language
 - Features: Bi-directional learning, recurrent memory, permutation modelling
- UAT + Layers XLNet-Base Details Addressing Look-ahead Bias
- ⇒ “**Good**” signal based on FOMC statement text within given window length
- ⇒ “Jointly” estimate optimal window **and** “good” signal

Universal Approximation Theorem

[Back to This Paper](#)[Back to Approach](#)[Back to NN Training Overview](#)

- ▶ Universal Approximation Theorem (Hornik et al., 1989; and others...) from ML literature:
 - Neural networks with **at least 1 hidden layer** can approximate **any** function
 - Existence theorem → Nothing about finding structure and training
- ▶ In reality, adding more layers:
 - ↓ number of parameters for each node function
 - ↓ computational, data, and training requirements

Details about XLNet-Base from Yang et al. (2019)

[Hyperparameters](#)[Back to Approach](#)

- ▶ **Overview:** Open-source, pretrained NN for text analysis
 - Paper version: XLNet-Base
- ▶ **Design:** Permutation- and autoregressive-based learning
 - 12 hidden layers (each of size 768)
 - 12 self-attention heads (each of size 64)
 - Vocabulary size of 32,000 word tokens
 - 110 million network parameters
- ▶ **Text Input:** Sequence of numerical vectors representing words and document
- ▶ **Transfer Learning:** “pretrained” parameters = Starting point for new task
 - SOTA status in translation, Q&A, word prediction, classification/regression
- ▶ **Initial Task:** Predict words using all permutations of text
- ▶ **Initial Data:** BookCorpus (11,038 books), English Wikipedia (6 mil articles)
 - Additional data for original, larger-scale NN:
 - Giga5 (9.9 mil news articles), ClueWeb12 (733 mil web pages), Common Crawl (1K+ TB text from web pages)

Hyperparameters for Fine-tuning XLNet-Base (1/2)

XLNet-Base Details

Hyperparameter	Value
Number of Layers	12
Hidden Size	768
Number of Attention Heads	12
Attention Head Size	64
Hidden Dropout	0.1
Attention Dropout	0.1
Max Sequence Length	512
Vocab Size	32,000

Table 3: The symbol “*” denotes the hyperparameter that undergoes tuning during the fine-tuning process. The random seed is set as a constant to ensure replicability by always initialising random components (e.g., weights and biases) to the same value. (cont.)

Hyperparameters for Fine-tuning XLNet-Base (2/2)

XLNet-Base Details

Hyperparameter	Value
Training Batch Size	8
Evaluation Batch Size	8
Max Num of Steps	2040
Max Num of Epochs	120
Manual Random Seed	47
Learning Rate	*
Warmup Ratio	0.06
Adam Epsilon	1e-8
Learning Rate Decay	Linear

Table 4: The symbol “*” denotes the hyperparameter that undergoes tuning during the fine-tuning process. The random seed is set as a constant to ensure replicability by always initialising random components (e.g., weights and biases) to the same value.

Addressing Look-ahead Bias

[Back to Approach](#)[Back to Loop](#)

- ▶ **Sarkar and Vafa (2024)**: NNs predict values in past using info. in the future.
 - NNs for text analysis trained with large amounts of data
 - High probability of future info. used in initial training of NN weights
 - Look-ahead bias addressed for 2 reasons:
 1. XLNet-Base initially trained **only** with BookCorpus and English Wikipedia
 - Very low probability of XLNet-Base initially trained on FOMC statements and futures data
 2. Pre-processed FOMC statements have no references to relevant times t and $t + 1$

[XLNet-Base Details](#)

Estimating Optimal Event Windows: Prepare for Train + Eval XLNet

[Back](#)

- ▶ **Goal:** “Good” signal from XLNet-Base for every FOMC statements
 - Method from ML literature: Train XLNet-Base on **fold**s of data

Estimating Optimal Event Windows: Prepare for Train + Eval XLNet

Back

- ▶ **Goal:** “Good” signal from XLNet-Base for every FOMC statements
 - Method from ML literature: Train XLNet-Base on **folds** of data
- ▶ Split data into training (80%) and testing (20%) subsamples (folds) **5 times**:
 - By **stratified sampling** [Why Stratified?](#) [Why CV?](#)
 - Conditions: Word count (200-word bins), FFR decision, FOMC chair, pre/post 2007
 - Every testing subsample share **NO** FOMC statements

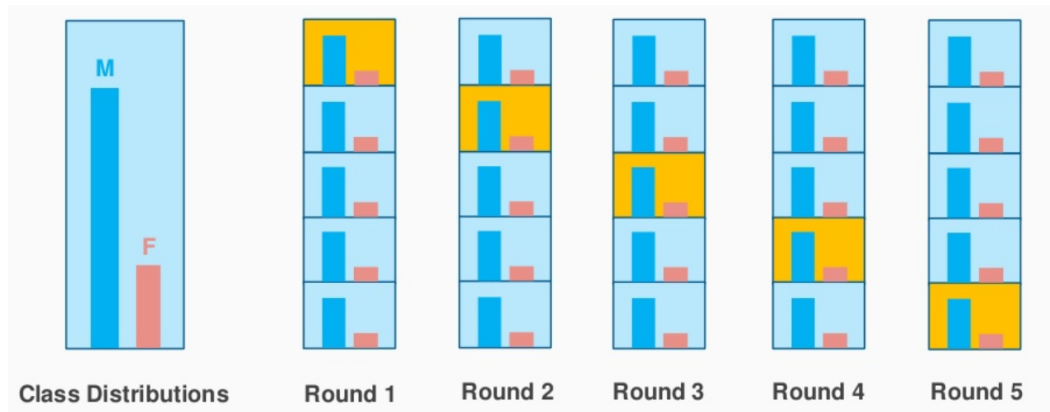
Estimating Optimal Event Windows: Prepare for Train + Eval XLNet

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 - By **stratified sampling** [Why Stratified?](#) [Why CV?](#)
 - Conditions: Word count (200-word bins), FFR decision, FOMC chair, pre/post 2007
 - Every testing subsample share **NO** FOMC statements
- ▶ **Result:** XLNet-Base learns $f(\text{FOMC statement text}) = DP_{t+n}$ for each fold:
 - With **equal distribution** of FOMC statements based on characteristics

Estimating Optimal Event Windows: Stratified CV Visual

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Estimating Optimal Event Windows: Method “Loop”

[Back to Method](#)

For each interest-rate and equity futures contract:

Estimating Optimal Event Windows: Method “Loop”

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For each interest-rate and equity futures contract:

- ▶ For each DP_{t+n} up to $t + 60$:

Estimating Optimal Event Windows: Method “Loop”

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For each interest-rate and equity futures contract:

- ▶ For each DP_{t+n} up to $t + 60$:
 - For each fold:
 1. Fine-tune NN parameters and hyperparameters to fit training data
[NN Training Overview](#) [Hyperparameter Tuning](#) [Addressing Look-ahead Bias](#)
 2. Evaluate NN on testing data → Choose hyperparameters that yield highest R^2_{OOS}

Estimating Optimal Event Windows: Method “Loop”

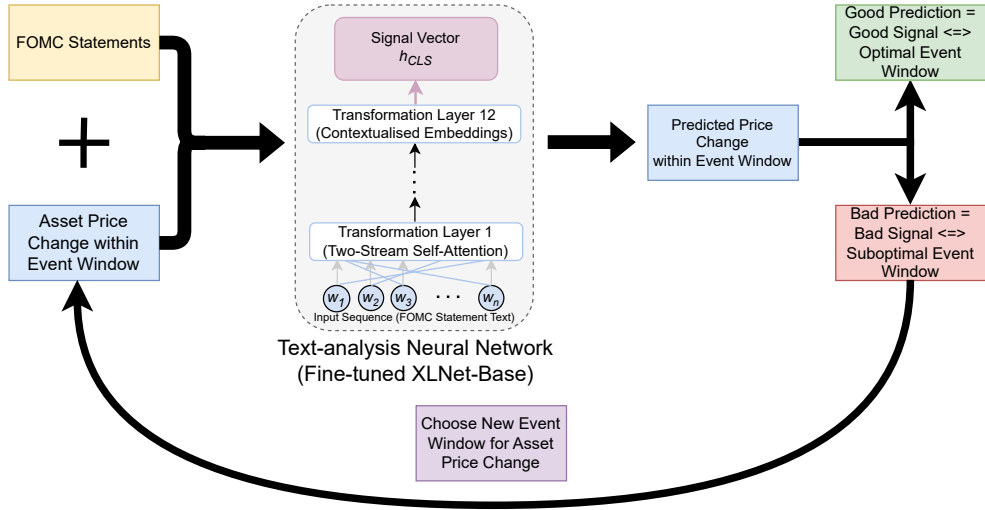
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For each interest-rate and equity futures contract:

- ▶ For each DP_{t+n} up to $t + 60$:
 - For each fold:
 1. Fine-tune NN parameters and **hyperparameters** to fit training data
[NN Training Overview](#) [Hyperparameter Tuning](#) [Addressing Look-ahead Bias](#)
 2. Evaluate NN on testing data → Choose hyperparameters that yield highest R_{OOS}^2
 3. **Final Output:** $\overline{R_{OOS}^2} := R_{OOS}^2$ **averaged across 5 folds**
 - Other R_{OOS}^2 metrics: Min, max, 75th, 25th percentiles

Estimating Optimal Event Windows: Method Visual

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NN Training Overview

[Back to Loop](#)

- ▶ Train NN → Fine-tune parameters and hyperparameters to fit training data

[Small NN Ex](#)

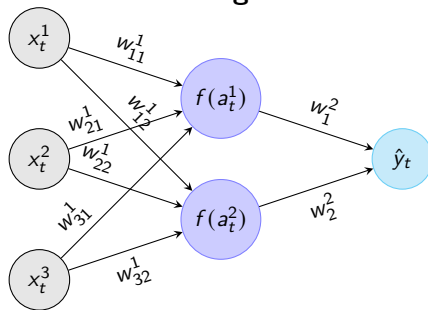
1. Fix network structure (layers and nodes) + non-tuned hyperparameters [UAT + Layers](#)
2. Choose value for hyperparameter that will be tuned [Hyperparameter Tuning](#)
3. Iteratively update parameters to $\downarrow \widehat{MSE}_{IS}$
4. Evaluate NN → Judge based on R^2_{OOS}
5. Poor performance → Go back to step 1

Small NN Example

[Back to NN Training Overview](#)

- ▶ **Data:** 4 variables x_t^1, x_t^2, x_t^3, y_t
- ▶ **Goal:** Predict y_t from $X \equiv x_t^1, x_t^2, x_t^3$
- ▶ **Example:** 2 layers, 2 “hidden” nodes
- ▶ From X_t to $\hat{y}_t$ for observation $t \in T$:
 - Linearly combine $x_t^1, x_t^2, x_t^3 \rightarrow a_t^1, a_t^2$
 - f is a non-linear function
 - $\hat{y}_t$ is predicted output
- ▶ **Training** prediction error $\rightarrow$ update weights w
- ▶ **Testing** prediction error $\rightarrow$ update network structure

NN Figure



NN Matrix Algebra

$$\begin{bmatrix} x_t^1 & x_t^2 & x_t^3 \end{bmatrix} \begin{bmatrix} w_{11}^1 & w_{12}^1 \\ w_{21}^1 & w_{22}^1 \\ w_{31}^1 & w_{32}^1 \end{bmatrix} = \begin{bmatrix} a_t^1 & a_t^2 \end{bmatrix}$$

$$\begin{bmatrix} f(a_t^1) & f(a_t^2) \end{bmatrix} \begin{bmatrix} w_1^2 \\ w_2^2 \end{bmatrix} = \hat{y}_t$$

Why Stratified Sampling?

[Back to CV](#)

► Why stratified over random splitting?

1. Transfer learning → Lower data requirements for NNs **BUT**
2. Large sample size for NNs → Fold $\approx$ Population for characteristics
 - Can use random k -fold cross validation
3. Finite sample size for NNs $\rightarrow$ Fold $\approx$ Population
 - Create folds **conditioned on class dist** can help
4. Minimises diff between pop and fold distributions of FOMC statement characteristics
5. **Result:** Better learning and predictive performance from NN

Estimating Optimal Event Windows: Accuracy Metrics

[Back to Method](#)[Summary Text](#)

- ▶ For each fold, primary metric to judge NN = **generalised $R^2 := R^2_{OOS}$**
- ▶ Make adjustments from typical definition because:

[R² Details](#)

Estimating Optimal Event Windows: Accuracy Metrics

[Back to Method](#)[Summary Text](#)

- ▶ For each fold, primary metric to judge NN = generalised $R^2 := R^2_{OOS}$
- ▶ Make adjustments from typical definition because:
 1. NN is a non-linear regression $\implies \rho^2 \neq R^2$

[R² Details](#)

Estimating Optimal Event Windows: Accuracy Metrics

[Back to Method](#)[Summary Text](#)

- ▶ For each fold, primary metric to judge NN = generalised $R^2 := R^2_{OOS}$
- ▶ Make adjustments from typical definition because:
 1. NN is a non-linear regression $\implies \rho^2 \neq R^2$
 2. Judging out-of-sample (OOS) performance, not in-sample

[R² Details](#)

Estimating Optimal Event Windows: Accuracy Metrics

[Back to Method](#)[Summary Text](#)

- ▶ For each fold, primary metric to judge NN = **generalised $R^2 := R^2_{OOS}$**
- ▶ Make adjustments from typical definition because:
 1. NN is a non-linear regression $\implies \rho^2 \neq R^2$
 2. Judging **out-of-sample (OOS)** performance, not in-sample
- ▶ **Other Tracked Metrics:** ρ_{OOS} , $\widehat{MAE_{OOS}}$, $\widehat{MSE_{IS}}$

[R² Details](#)

Estimating Optimal Event Windows: Accuracy Metrics

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- ▶ For each fold, primary metric to judge NN from Hawinkel et al. (2024):

$$R_{OOS}^2 = 1 - \frac{\widehat{MSE}}{\widehat{MST}} = 1 - \frac{T^{-1} \sum_{i=1}^T (y_i - \hat{y}_i)^2}{\frac{T+1}{T(T-1)} \sum_{i=1}^T (y_i - \overline{yIS})^2}, \quad (8)$$

Estimating Optimal Event Windows: Accuracy Metrics

[Back](#)

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$$R_{OOS}^2 = 1 - \frac{\widehat{MSE}}{\widehat{MST}} = 1 - \frac{T^{-1} \sum_{i=1}^T (y_i - \hat{y}_i)^2}{\frac{T+1}{T(T-1)} \sum_{i=1}^T (y_i - \overline{yIS})^2}, \quad (8)$$

- ▶ **Definition:** Comparison between two models: NN and **null model**
 - Null model: $\overline{yIS}$ as prediction

Estimating Optimal Event Windows: Accuracy Metrics

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- ▶ **Definition:** Comparison between two models: NN and **null model**
 - **Null model:** $\overline{yIS}$ as prediction
- ▶ **Interpretation:** % of null model's MSE explained by NN
 - **NOT** % of DP_{t+n} variance explained by NN $\because$ nonlinearity

Estimating Optimal Event Windows: Accuracy Metrics

[Back](#)

- ▶ For each fold, primary metric to judge NN from Hawinkel et al. (2024):

$$R_{OOS}^2 = 1 - \frac{\widehat{MSE}}{\widehat{MST}} = 1 - \frac{T^{-1} \sum_{i=1}^T (y_i - \hat{y}_i)^2}{\frac{T+1}{T(T-1)} \sum_{i=1}^T (y_i - \overline{yIS})^2}, \quad (8)$$

- ▶ **Definition:** Comparison between two models: NN and **null model**

- **Null model:** $\overline{yIS}$ as prediction

- ▶ **Interpretation:** % of null model's MSE explained by NN

- **NOT** % of DP_{t+n} variance explained by NN $\because$ nonlinearity

- ▶ **Explicit objective function:** Minimise $\widehat{MSE}$ during fine-tuning

- $\min \widehat{MSE} = \max R_{OOS}^2$

Estimating Optimal Event Windows: Accuracy Metrics

[Back](#)

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- ▶ **Explicit objective function:** Minimise $\widehat{MSE}$ during fine-tuning

- $\min \widehat{MSE} = \max R_{OOS}^2$

- ▶ **Other tracked metrics:** ρ_{OOS} , $\widehat{MAE}_{OOS}$, $\widehat{MSE}_{IS}$

Why Cross Validation?

[Back to Stratified CV](#)

- ▶ Purpose: See how well model performs on unseen data while addressing overfitting
 - Also for assessing prediction stability across folds
- ▶ Popular usage in ML literature: Model selection
- ▶ **One Model:** Reduce prediction variation coming from folds themselves
 1. Allows model to predict for **all** sample observations
 2. Some folds might be ↑ “lucky” than others

Tuning XLNet-Base Hyperparameters

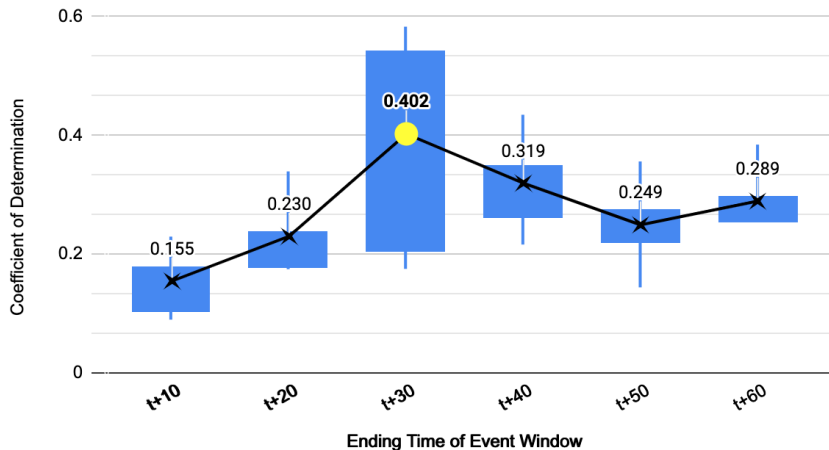
[Back to Loop](#)

- ▶ Hyperparameters: Variables outside NN structure that affect training performance
 - “Tuned” hyperparameter: **Learning rate**
 - Analogy: CPU & GPU overclocking (e.g., voltage, temperature, clock frequencies)
- ▶ For each 10-minute interval, “tune” learning rate of XLNet:
 - “Best chance” of approximating $f(\text{FOMC statement text}) = DP_{t+n}$
 - Tuning process takes **1 computation day** for each DP_{t+n}

Optimal Event Windows: *FF3*

[Summary Visual](#) [Summary Text](#)

Out-of-sample R^2 for FF3 (Averaged Across Splits)

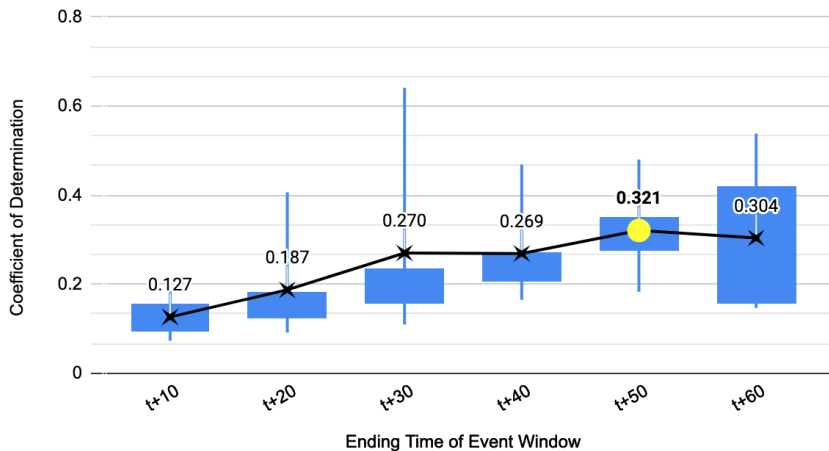


Optimal Event Windows: *USc2*

Summary Visual

Summary Text

Out-of-sample R^2 for USc2 (Averaged Across Splits)



Optimal Event Windows: R^2_{OOS} Table (1/2)[Summary Text](#) [Summary Visual \(Intro\)](#)

Asset	$\overline{R^2_{OOS}}$, 30-min	$\overline{R^2_{OOS}}$, Optimal	Difference
<i>FF1</i>	35.0%	37.2%	+2.2 p.p.
<i>FF2</i>	28.7%	34.5%	+5.8 p.p.
<i>FF3</i>	23.0%	40.2%	+17.2 p.p.
<i>FF4</i>	29.8%	43.3%	+13.5 p.p.
<i>EDcm2</i>	18.3%	23.3%	+5 p.p.
<i>EDcm3</i>	14.0%	18.2%	+4.2 p.p.
<i>EDcm4</i>	11.2%	16.0%	+4.8 p.p.
<i>TUc1</i>	21.3%	24.4%	+3.1 p.p.
<i>TUc2</i>	16.5%	19.4%	+2.9 p.p.

Table 5: Differences of $\overline{R^2_{OOS}}$ between 30-minute and Optimal Event Windows (cont.)

Optimal Event Windows: R^2_{OOS} Table (2/2)

Summary Text Summary Visual (Intro)

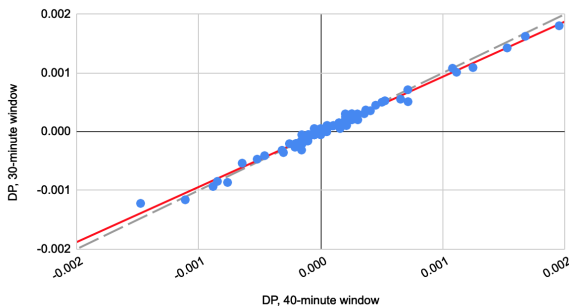
Asset	R^2_{OOS} , 30-min	R^2_{OOS} , Optimal	Difference
<i>FVc1</i>	11.7%	21.4%	+9.7 p.p.
<i>FVc2</i>	12.7%	19.2%	+6.5 p.p.
<i>TYc1</i>	11.7%	25.5%	+13.8 p.p.
<i>TYc2</i>	11.4%	23.9%	+12.5 p.p.
<i>USc1</i>	15.7%	28.5%	+12.8 p.p.
<i>USc2</i>	18.7%	32.1%	+13.4 p.p.
<i>SPX</i>	18.4%	23.2%	+4.8 p.p.
<i>ESc1</i>	22.9%	27.7%	+4.8 p.p.
<i>ESc2</i>	19.3%	23.5%	+4.2 p.p.

Table 6: Differences of R^2_{OOS} between 30-minute and Optimal Event Windows

Optimal Windows: Diff Windows, Diff Responses ($FF3^{***}$)

[Other Assets](#)[Summary Text](#)

Market Responses in Different Event Windows (FF3)

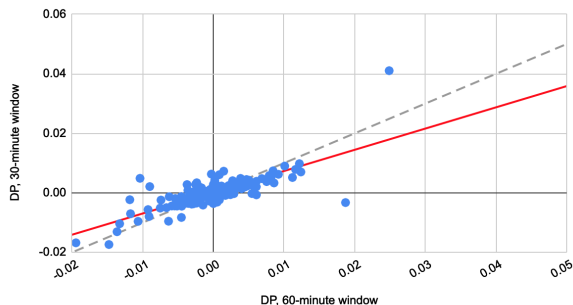


- ▶ **Takeaway:** On average, markets **under-react**, ex-post, to FOMC statement text
- ▶ **“Soft”** information = Longer to process → Info asymmetry to resolve
 - Post-FOMC drift: Indriawan et al. (2021); Brooks et al. (2023)
 - Fed Info/RTN Effect: Nakamura and Steinsson (2018); Bauer and Swanson (2023b)
 - Investor heterogeneity/Sequential learning: Ai et al. (2021)
 - Macro attention/Risk premia: Fisher et al. (2022)

Optimal Windows: Diff Windows, Diff Responses ($USc2^{***}$)

[Other Assets](#)[Summary Text](#)

Market Responses in Different Event Windows (USc2)

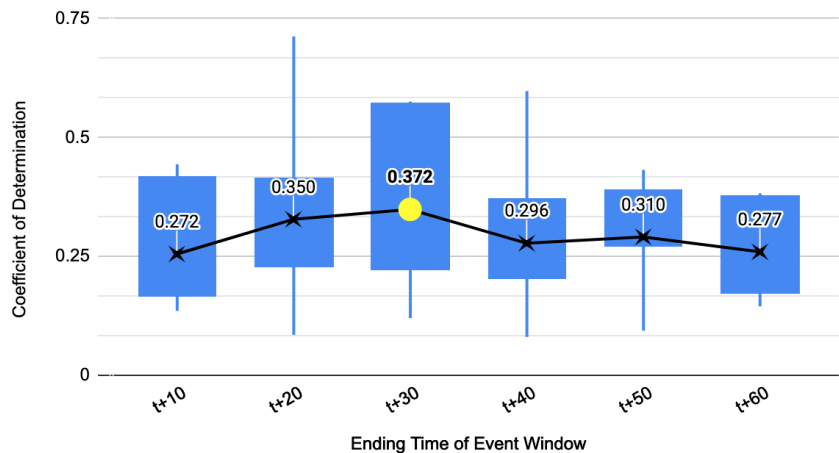


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 - Investor heterogeneity/Sequential learning: Ai et al. (2021)
 - Macro attention/Risk premia: Fisher et al. (2022)

Optimal Event Windows: *FF1*

[Back to Summary Text](#)

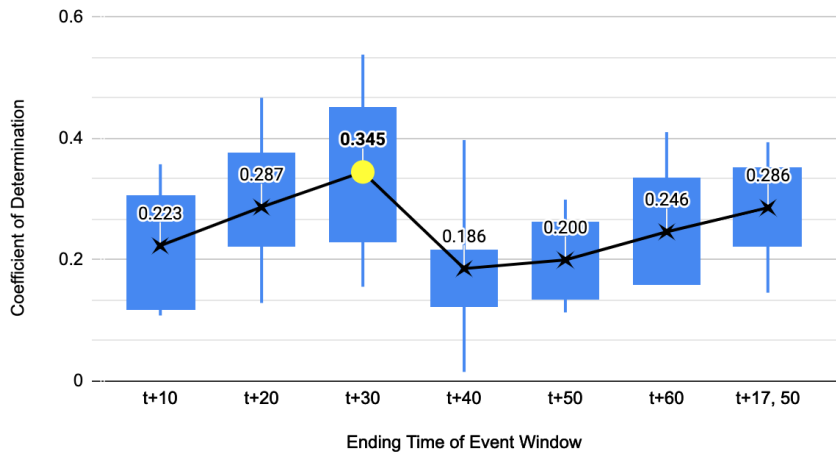
Out-of-sample R^2 for FF1 (Averaged Across Splits)



Optimal Event Windows: *FF2*

[Back to Summary Text](#)
[One Signal](#)

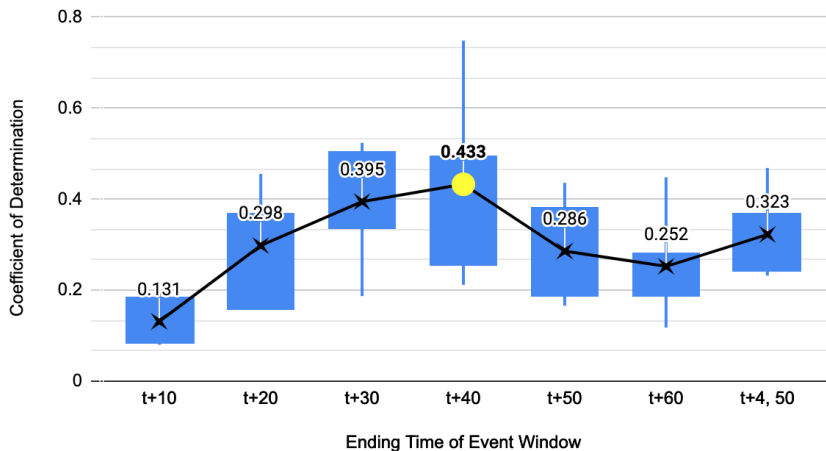
Out-of-sample R^2 for FF2 (Averaged Across Splits)



Optimal Event Windows: *FF4*

[Back to Summary Text](#)
[One Signal](#)

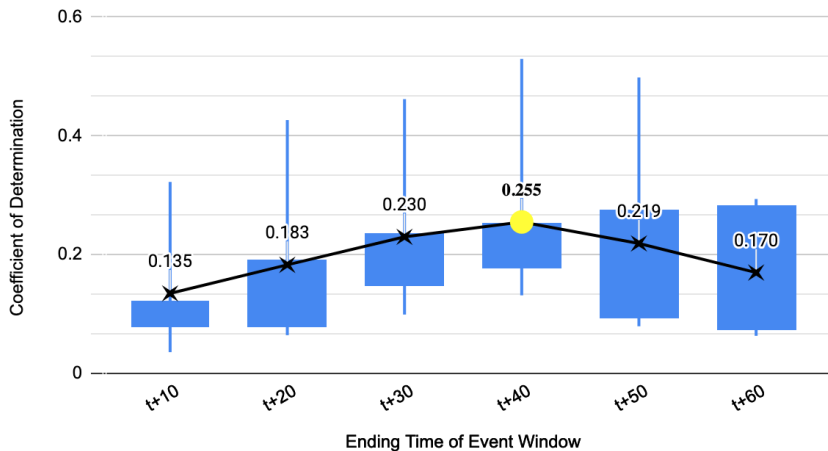
Out-of-sample R^2 for FF4 (Averaged Across Splits)



Optimal Event Windows: ED_{cm2}

[Back to Summary Text](#)

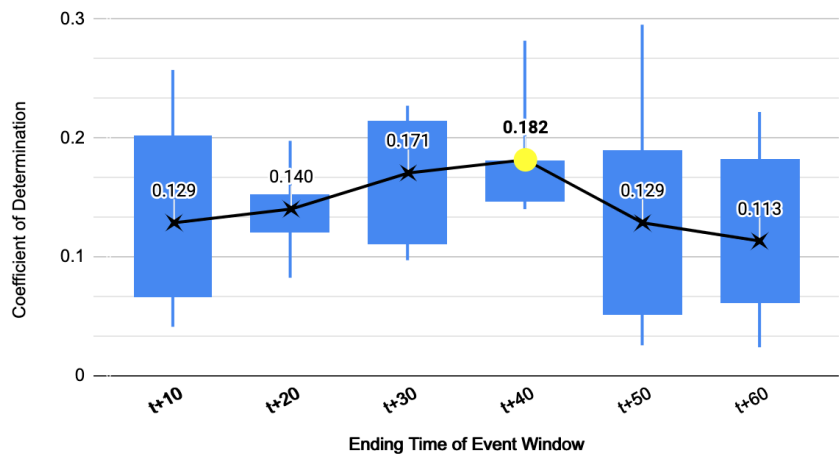
Out-of-sample R^2 for ED_{cm2} (Averaged Across Splits)



Optimal Event Windows: *EDcm3*

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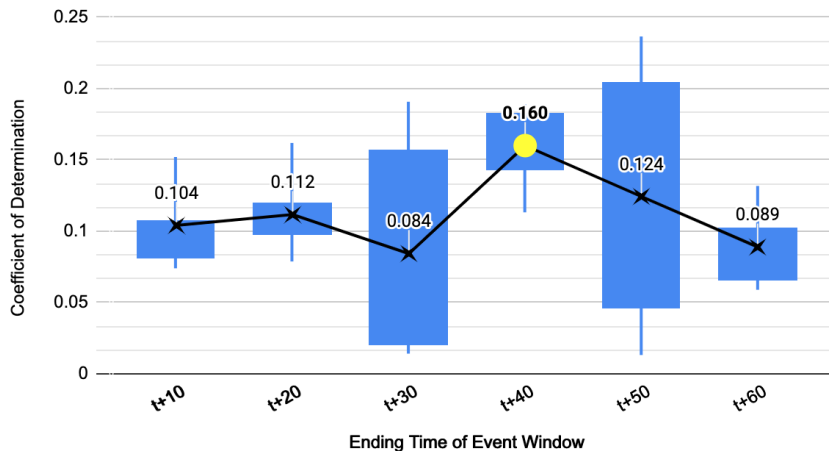
Out-of-sample R^2 for *EDcm3* (Averaged Across Splits)



Optimal Event Windows: *EDcm4*

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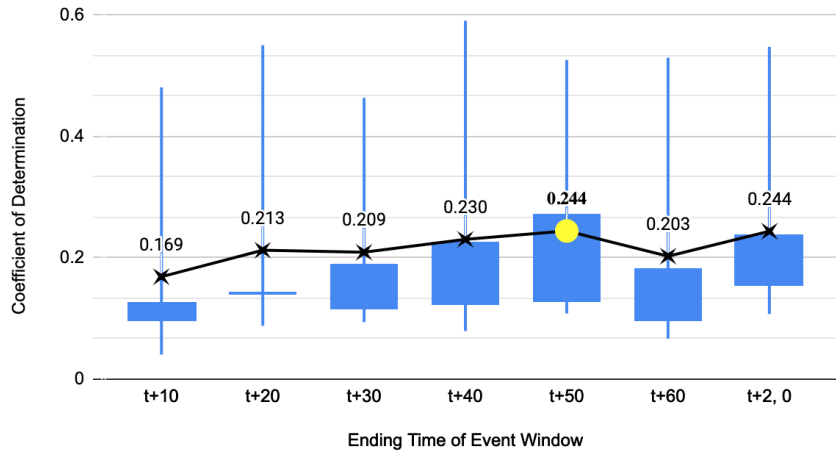
Out-of-sample R^2 for *EDcm4* (Averaged Across Splits)



Optimal Event Windows: TU_{c1}

[Back to Summary Text](#)
[One Signal](#)

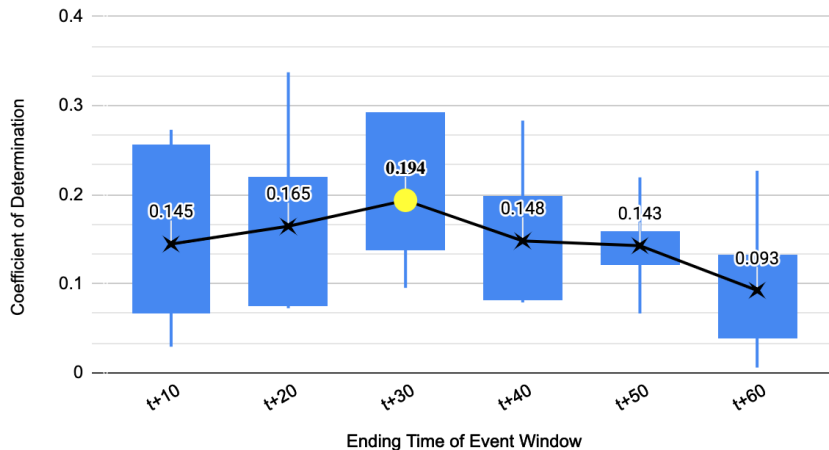
Out-of-sample R^2 for TU_{c1} (Averaged Across Splits)



Optimal Event Windows: $TUc2$

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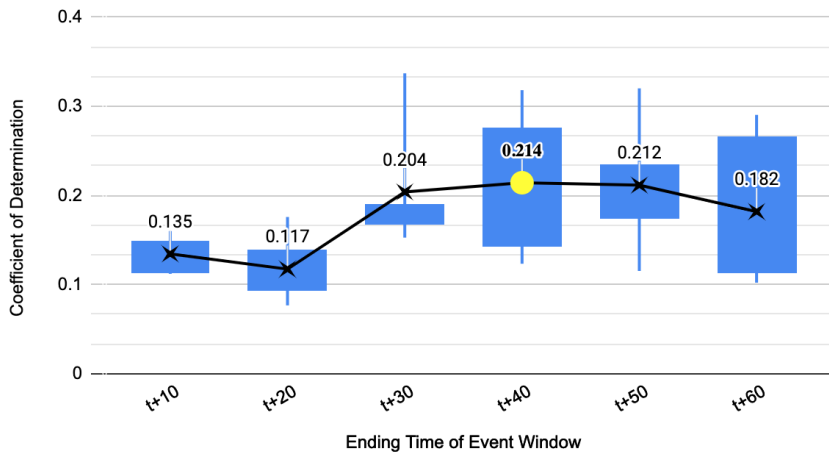
Out-of-sample R^2 for $TUc2$ (Averaged Across Splits)



Optimal Event Windows: $FVc1$

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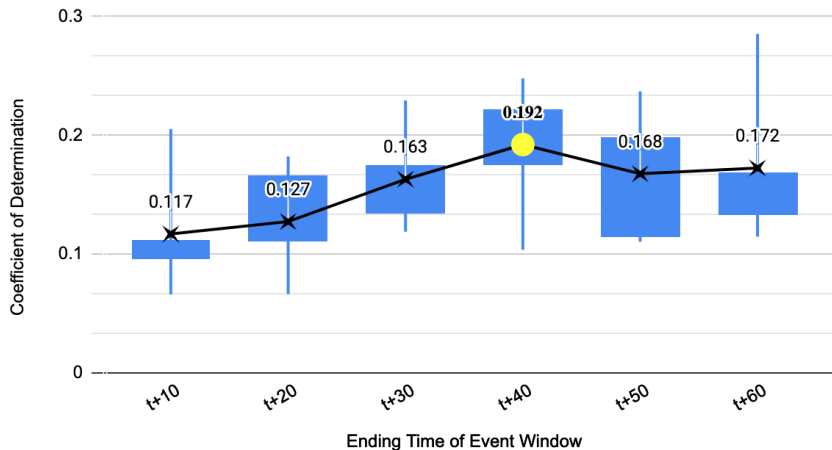
Out-of-sample R^2 for $FVc1$ (Averaged Across Splits)



Optimal Event Windows: $FVc2$

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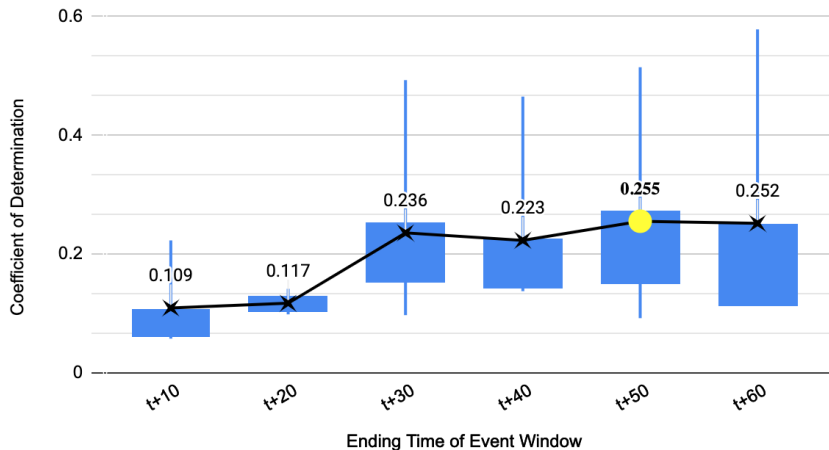
Out-of-sample R^2 for $FVc2$ (Averaged Across Splits)



Optimal Event Windows: TY_{c1}

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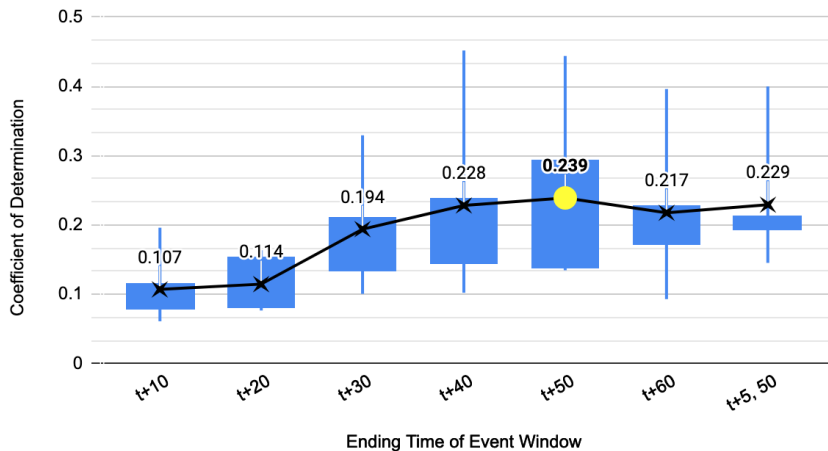
Out-of-sample R^2 for TY_{c1} (Averaged Across Splits)



Estimating Optimal Event Windows: $TYc2$

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[One Signal](#)

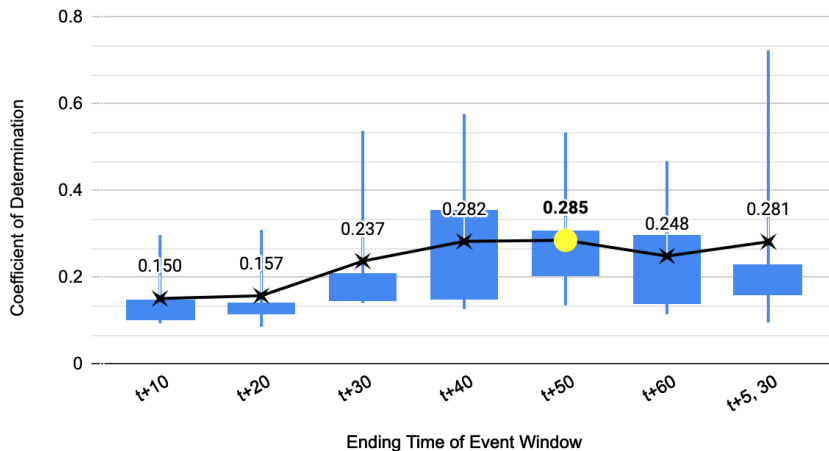
Out-of-sample R^2 for $TYc2$ (Averaged Across Splits)



Optimal Event Windows: *USc1*

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[One Signal](#)

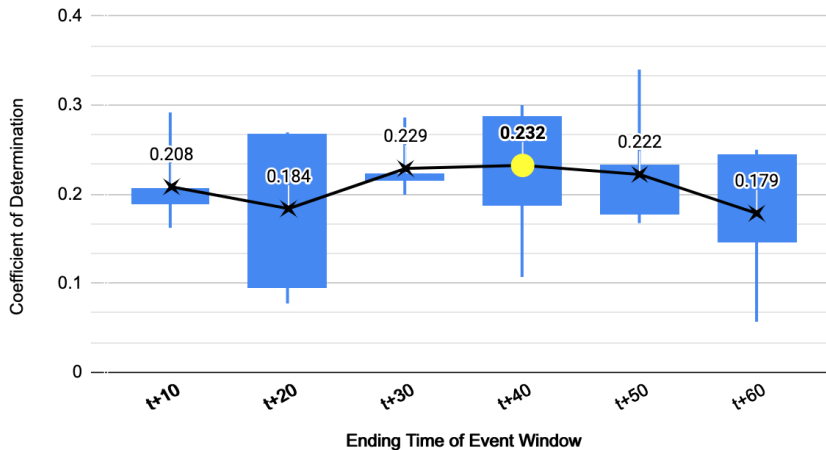
Out-of-sample R^2 for *USc1* (Averaged Across Splits)



Optimal Event Windows: *SPX*

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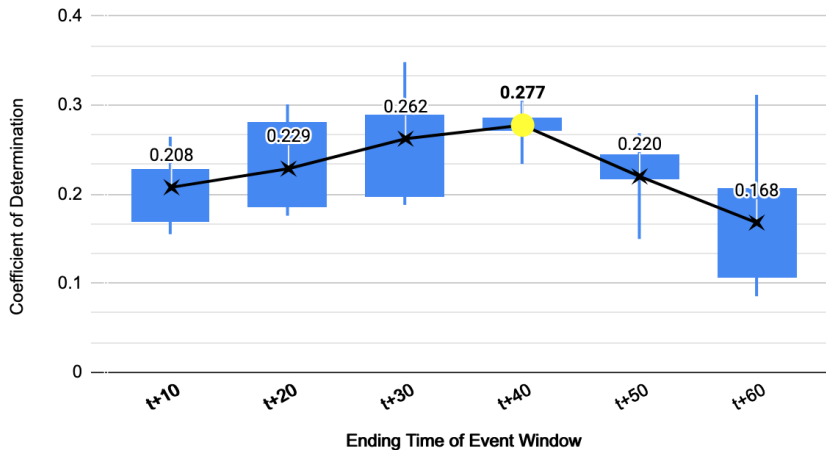
Out-of-sample R^2 for SPX (Averaged Across Splits)



Optimal Event Windows: $ESc1$

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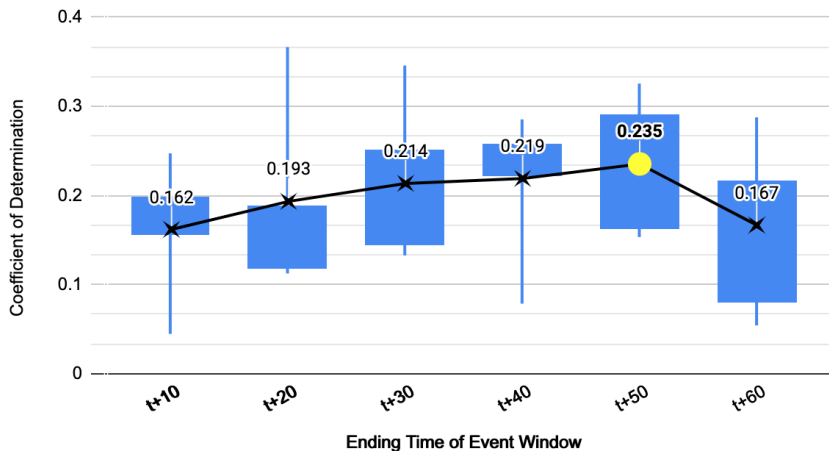
Out-of-sample R^2 for $ESc1$ (Averaged Across Splits)



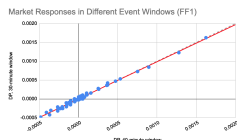
Optimal Event Windows: *ESc2*

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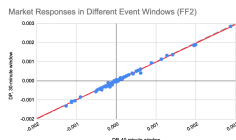
Out-of-sample R^2 for *ESc2* (Averaged Across Splits)



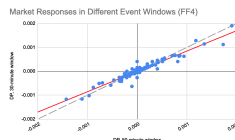
Optimal Event Windows: Diff Windows, Diff Responses (1/2)



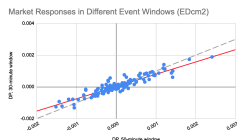
(a) $FF1^{**}$



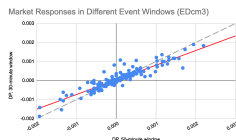
(b) $FF2^{***}$



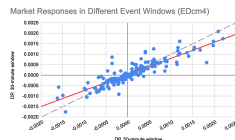
(c) $FF4^{***}$



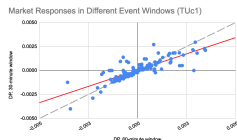
(d) $EDcm2^{***}$



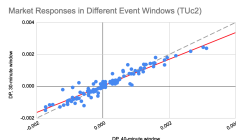
(e) $EDcm3^{***}$



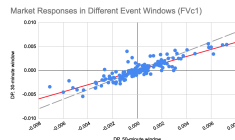
(f) $EDcm4^{***}$



(g) $TUC1^{***}$

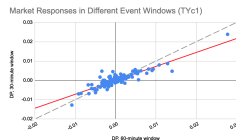
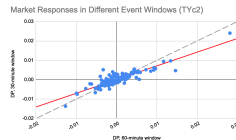
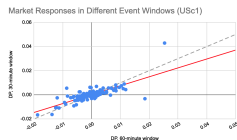
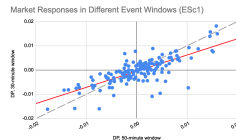


(h) $TUC2^{***}$



(i) $FVC1^{***}$

Optimal Event Windows: Diff Windows, Diff Responses (2/2)

(a) $FVc2^{***}$ (b) $TYc1^{***}$ (c) $TYc2^{***}$ (d) $USc1^{***}$ (e) SPX^{***} (f) $ESc1^{***}$ (g) $ESc2^{***}$

Interest-rate Futures Prices into MP Surprises: $mp1$

[Back to MP surprises](#)

- ▶ For given expiry month, FFF pays out, on last day, $100 - \text{avg FFR}$
- ▶ Price of $(1 - j)$ month-ahead FFF at time t for FOMC meeting τ : $p_{\tau,t}^{FFj}$
- ▶ Expected avg FFR at t for τ : $FFj_{\tau,t} = 100 - p_{\tau,t}^{FFj}$
- Current-meeting FFR surprise $mp1_{\tau,t+n}$:

$$mp1_{\tau,t+n} = \frac{m}{m-d} (FF1_{\tau,t+n} - FF1_{\tau,t-10}), \quad (9)$$

- ▶ Day d of month, days m in month
- ▶ If $m - d + 1 \leq 7 \implies mp1_{\tau,t+n} = FF2_{\tau,t+n} - FF2_{\tau,t-10}$
- ▶ **Futures Contracts:** $FF1, FF2$

Interest-rate Futures Prices into MP Surprises: $mp2$

[Back to MP surprises](#)

- ▶ Number of months out $(j - 1)$ containing next meeting
- ▶ Next-meeting FFR surprise $mp2_{\tau,t+n}$:

$$mp2_{\tau,t+n} = \frac{m_2}{m_2 - d_2} \left\{ [FFj_{\tau,t+n} - FFj_{\tau,t-10}] - \frac{d_2}{m_2} mp1_{\tau,t+n} \right\}, \quad (10)$$

- ▶ Day d_2 of next-meeting month, days m_2 in next-meeting month
- ▶ If $m_2 - d_2 + 1 \leq 7 \implies mp2_{\tau,t+n} = FF(j+1)_{\tau,t+n} - FF(j+1)_{\tau,t-10}$
- ▶ **Futures Contracts:** $FF2, FF3, FF4$

Interest-rate Futures Prices into MP Surprises: Δedj

[Back to MP surprises](#)

- ▶ On last day of last quarter, ED pays out 100– 3-month US dollar BBA LIBOR rate
- ▶ Price at time t of j^{th} nearest quarterly ED contract for meeting τ : $p_{\tau,t}^{edj}$
- ▶ Implied rate at t for τ : $edj_{\tau,t} = 100 - p_{\tau,t}^{edj}$
- Implied rate surprise j -quarters out $edj_{\tau,t+n}$:

$$edj_{\tau,t+n} = edj_{\tau,t+n} - edj_{\tau,t-10}, \quad (11)$$

- ▶ Day d of month, days m in month
- ▶ **Futures Contracts:** $EDcm2, EDcm3, EDcm4$

Interest-rate Futures Prices into MP Surprises: Δtk

[Back to MP surprises](#)

- ▶ On expiry quarter, Tsy futures obliges seller to deliver bond within maturities range
- ▶ Price at time t of j^{th} nearest quarterly k -year Treasury contract for meeting τ : $p_{\tau,t}^{tkj}$
- Implied yield surprise for meeting τ $tk_{\tau,t+n}$:

$$tk_{\tau,t+n} = - \left(p_{\tau,t+n}^{tkj} - p_{\tau,t-10}^{tkj} \right) / l, \quad (12)$$

- ▶ If $\tau \in \{\text{Mar, Jun, Sep, Dec}\} \implies tk_{\tau,t+n} = - \left(p_{\tau,t+n}^{tkj+1} - p_{\tau,t-10}^{tkj+1} \right) / l$
- ▶ Approximated maturities $l \in \{2, 4, 7, 15\}$ by Gürkaynak, Kisacikoğlu, et al. (2020)
- ▶ **Futures Contracts:** $TUc1, TUc2; FVc1, FVc2; TYc1, TYc2; USc1, USc2$

Principal Component Analysis

[Back to MP Shocks](#)

- ▶ **Purpose:** Reduces dimensionality without sacrificing data variation
- ▶ **Example:** Variables x^1, x^2 ; N observations
- ▶ 1st Principal component:
$$\underbrace{PC1}_{N \times 1} = \underbrace{X}_{N \times 2} \cdot \underbrace{V}_{2 \times 1},$$
 1. X = Covariance matrix of variables
 2. V = Eigenvector of covariance matrix X that has largest eigenvalue
- ▶ Largest eigenvalue $\rightarrow$ Captures most common variation in data
- $\rightarrow$ Corresponding eigenvector is “direction” explaining data variation

Monetary Policy Shocks: Visual Diff from Window Choice (1/2)

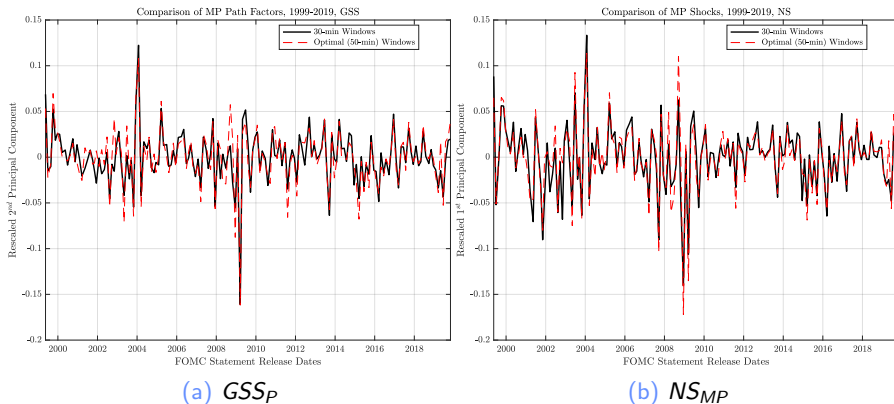
[Back to MP Shocks](#)


Figure 1: Comparing MP Shock Series Derived from Optimal Window Length v. 30 Minutes

Monetary Policy Shocks: Visual Diff from Window Choice (2/2)

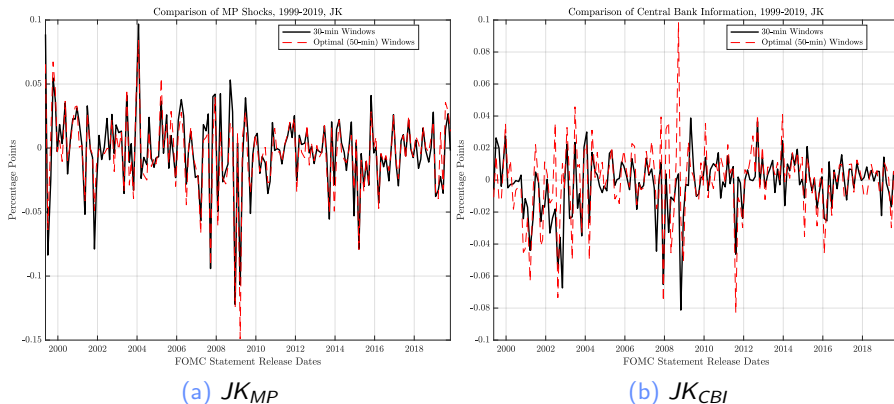
[Back to MP Shocks](#)


Figure 2: Comparing MP Shock Series Derived from Optimal Window Length v. 30 Minutes

Monetary Policy Shocks, Statement Frequency: Summary Table

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Metric	GSS_T	GSS_P	NS_{MP}	JK_{MP}	JK_{CBI}
Count	165 (165)	165 (165)	165 (165)	165 (165)	165 (165)
Mean	0 (0)	0 (0)	0 (0)	-0.0023 (-0.0035)	-0.0028 (-0.0004)
SD	0.0219 (0.0216)	0.0282 (0.0309)	0.0356 (0.0375)	0.0305 (0.0297)	0.0180 (0.0233)
Max	0.0756 (0.0652)	0.1223 (0.1079)	0.1333 (0.1135)	0.0968 (0.0840)	0.0388 (0.0984)
75 th	0.0076 (0.0084)	0.0134 (0.0167)	0.0193 (0.0203)	0.0175 (0.0124)	0.0069 (0.0112)
Median	0.0011 (0.0023)	-0.0010 (0.0019)	0.0033 (0.0003)	-0.0017 (-0.0009)	-0.0003 (0.0022)
25 th	-0.0069 (-0.0077)	-0.0124 (-0.0127)	-0.0187 (-0.0164)	-0.0153 (-0.0138)	-0.0086 (-0.0103)
Min	-0.1038 (-0.1025)	-0.1611 (-0.1618)	-0.1403 (-0.1722)	-0.1222 (-0.1491)	-0.0812 (-0.0828)

Table 7: Descriptive Statistics for Monetary Policy Shock Series, FOMC Statement Frequency for 1999–2019

Impulse Response Variables: Summary Table

[Back to LP](#)

Metric	$\ln(CPI)$	$\ln(IP)$	EBP	TY_2
Count	246	246	246	246
Mean	5.356	4.597	0.110	2.210
SD	0.126	0.057	0.715	1.794
Max	5.551	4.706	3.283	6.650
75 th	5.466	4.642	0.341	3.569
Median	5.381	4.605	-0.108	1.673
25 th	5.241	4.548	-0.334	0.662
Min	5.112	4.467	-1.140	0.188

Table 8: Descriptive Statistics for Impulse Response Variables, Monthly for 1999–2019

Monetary Policy Shocks, Monthly Frequency: Summary Table

[Back to LP](#)

- Months without FOMC meeting → Shock values set to zero

Metric	GSS_T	GSS_P	NS_{MP}	JK_{MP}	JK_{CBI}
Count	165 (165)	165 (165)	165 (165)	165 (165)	165 (165)
Mean	0 (0)	0 (0)	0 (0)	-0.0015 (-0.0024)	-0.0019 (-0.0002)
SD	0.0180 (0.0177)	0.0230 (0.0253)	0.0291 (0.0307)	0.0250 (0.0243)	0.0148 (0.0190)
Max	0.0756 (0.0652)	0.1223 (0.1079)	0.1333 (0.1135)	0.0968 (0.0840)	0.0388 (0.0984)
75 th	0.0042 (0.0042)	0.0078 (0.0089)	0.0114 (0.0110)	0.0055 (0.0058)	0.0031 (0.0075)
Median	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
25 th	-0.0021 (-0.0011)	-0.0064 (-0.0052)	-0.0036 (-0.0039)	-0.0073 (-0.0070)	-0.0037 (-0.0035)
Min	-0.1038 (-0.1025)	-0.1611 (-0.1618)	-0.1403 (-0.1722)	-0.1222 (-0.1491)	-0.0812 (-0.0828)

Table 9: Descriptive Statistics for Monetary Policy Shock Series, Monthly Frequency for 1999–2019

MP Shocks: Nominal Interest Rates

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	30-minute Window				Optimal Window				Difference		
	ΔTY_1	ΔTY_2	ΔTY_5	ΔTY_{10}	ΔTY_1	ΔTY_2	ΔTY_5	ΔTY_{10}	ΔTY_2	ΔTY_5	ΔTY_{10}
GSS_T	1.00*** (0.28)	0.82*** (0.34)	0.15 (0.43)	-0.37 (0.41)	1.00*** (0.23)	0.78*** (0.28)	0.08 (0.33)	-0.42 (0.32)	-0.04	-0.07	-0.05
GSS_P	1.00*** (0.09)	1.46*** (0.09)	1.89*** (0.25)	1.64*** (0.36)	1.00*** (0.11)	1.51*** (0.10)	1.93*** (0.21)	1.66*** (0.30)	+0.05	+0.04	+0.02
NS_{MP}	1.00*** (0.07)	1.24*** (0.09)	1.29*** (0.19)	0.95*** (0.25)	1.00*** (0.09)	1.30*** (0.10)	1.39*** (0.18)	1.06*** (0.24)	+0.06	+0.11	+0.11
JK_{MP}	1.00*** (0.11)	1.30*** (0.15)	1.39*** (0.28)	0.99*** (0.36)	1.00*** (0.12)	1.35*** (0.16)	1.52*** (0.32)	1.16*** (0.43)	+0.04	+0.13	+0.17
JK_{CBI}	1.00*** (0.25)	1.04*** (0.30)	1.00*** (0.31)	0.82*** (0.29)	1.00*** (0.23)	1.20*** (0.25)	1.14*** (0.27)	0.85*** (0.26)	+0.16	+0.14	+0.03

Table 10: Diff in Responses of Nominal Interest Rates to Shocks from Event Window Choice

→ Using optimal windows → Bigger effects for MP shocks about forward guidance

MP Shocks: Real Interest Rates

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	30-minute Window			Optimal Window			Difference		
	$\Delta TIPS_2$	$\Delta TIPS_5$	$\Delta TIPS_{10}$	$\Delta TIPS_2$	$\Delta TIPS_5$	$\Delta TIPS_{10}$	$\Delta TIPS_2$	$\Delta TIPS_5$	$\Delta TIPS_{10}$
GSS_T	-0.81 (1.44)	0.02 (0.51)	-0.19 (0.45)	-0.90 (1.57)	0.09 (0.46)	-0.16 (0.37)	-0.09	+0.07	+0.03
GSS_P	2.21*** (0.49)	1.96*** (0.46)	1.74*** (0.44)	2.20*** (0.37)	2.03*** (0.38)	1.75*** (0.37)	-0.00	+0.06	+0.01
NS_{MP}	1.17 (0.73)	1.29*** (0.36)	1.08*** (0.30)	1.31** (0.63)	1.47*** (0.31)	1.20*** (0.28)	+0.14	+0.18	+0.13
JK_{MP}	1.40* (0.83)	1.40*** (0.47)	1.15*** (0.41)	1.66*** (0.63)	1.64*** (0.49)	1.38*** (0.46)	+0.26	+0.24	+0.23
JK_{CBI}	0.51 (0.85)	0.99*** (0.33)	0.85*** (0.25)	0.60 (0.92)	1.13*** (0.33)	0.84*** (0.25)	+0.09	+0.14	-0.01

Table 11: Diff in Responses of Real Interest Rates to Shocks from Event Window Choice

→ Using optimal windows → Bigger effects for MP shocks about forward guidance

MP Shocks: Break-even Inflation

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	30-minute Window			Optimal Window			Difference		
	ΔBEI_2	ΔBEI_5	ΔBEI_{10}	ΔBEI_2	ΔBEI_5	ΔBEI_{10}	ΔBEI_2	ΔBEI_5	ΔBEI_{10}
GSS_T	1.63 (1.46)	0.13 (0.29)	-0.18 (0.17)	1.67 (1.58)	-0.01 (0.29)	-0.26 (0.17)	+0.05	-0.14	-0.08
GSS_P	-0.75 (0.46)	-0.08 (0.24)	-0.10 (0.12)	-0.69* (0.36)	-0.10 (0.23)	-0.09 (0.12)	+0.06	-0.02	+0.01
NS_{MP}	0.07 (0.70)	-0.01 (0.23)	-0.13 (0.12)	-0.01 (0.65)	-0.07 (0.23)	-0.14 (0.12)	-0.08	-0.07	-0.01
JK_{MP}	-0.09 (0.81)	-0.01 (0.29)	-0.17 (0.13)	-0.31 (0.61)	-0.12 (0.25)	-0.22** (0.11)	-0.22	-0.11	-0.05
JK_{CBI}	0.54 (0.76)	0.01 (0.29)	-0.02 (0.20)	0.60 (0.86)	0.01 (0.30)	0.02 (0.23)	+0.07	+0.00	+0.04

Table 12: Diff in Responses of Break-even Inflation to Shocks from Event Window Choice

→ Using optimal windows → Bigger effects for MP shocks about forward guidance

MP Shocks: Stock Prices

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	30-minute Window			Optimal Window			Percentage Difference		
	DP_{SPX}	DP_{ESc1}	DP_{ESc2}	DP_{SPX}	DP_{ESc1}	DP_{ESc2}	DP_{SPX}	DP_{ESc1}	DP_{ESc2}
GSS_T	-8.40** (2.71)	-8.83** (2.68)	-7.25** (2.78)	-7.39** (3.10)	-7.43** (3.15)	-7.34** (3.12)	-11.99%	-15.92%	+1.25%
GSS_P	-6.14** (1.81)	-6.27** (1.83)	-6.12** (1.76)	-6.85** (2.88)	-6.96** (2.91)	-7.63** (2.81)	+11.51%	+11.00%	+24.61%
NS_{MP}	-6.92** (1.32)	-7.15** (1.37)	-6.51** (1.31)	-7.00** (1.85)	-7.10** (1.89)	-7.55** (1.84)	+1.23%	-1.00%	+16.00%
JK_{MP}	-14.76** (0.81)	-15.08** (0.91)	-13.73** (0.94)	-17.46** (1.04)	-17.77** (1.08)	-17.30** (1.06)	+18.25%	+17.88%	+26.00%
JK_{CBI}	15.19** (2.29)	13.84** (2.35)	15.18** (2.39)	14.08** (2.11)	14.44** (2.14)	12.12** (2.07)	-7.36%	-4.90%	-12.43%

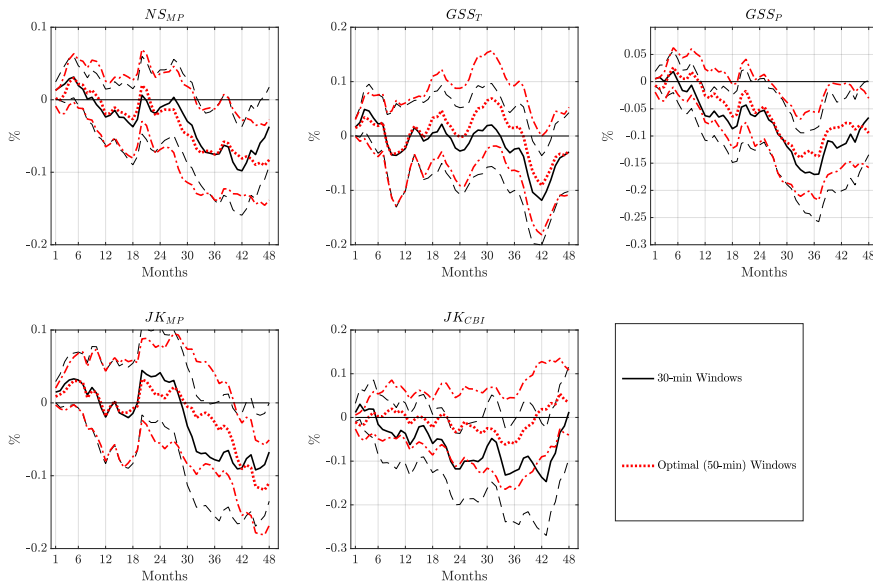
Table 13: Diff in Responses of Stock Prices to Shocks from Event Window Choice

Notes: Positive (negative) values → stronger (weaker) effect in same direction

→ Using optimal windows → Bigger effects for MP shocks about forward guidance

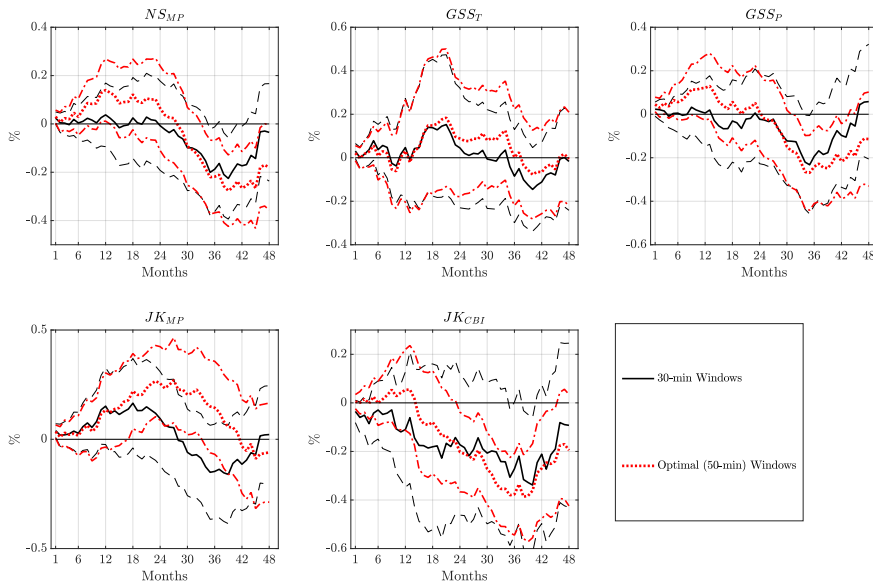
MP Shocks: Impulse Responses for CPI

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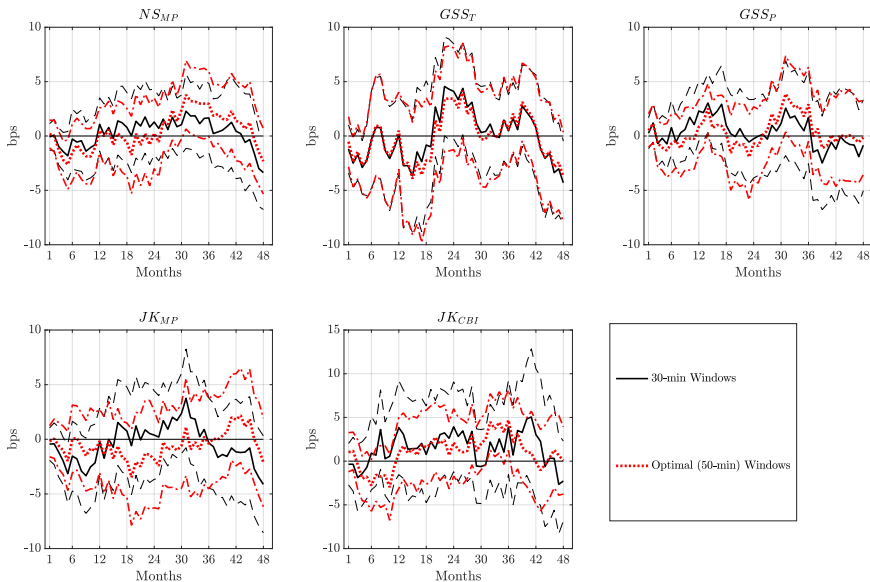
MP Shocks: Impulse Responses for IP

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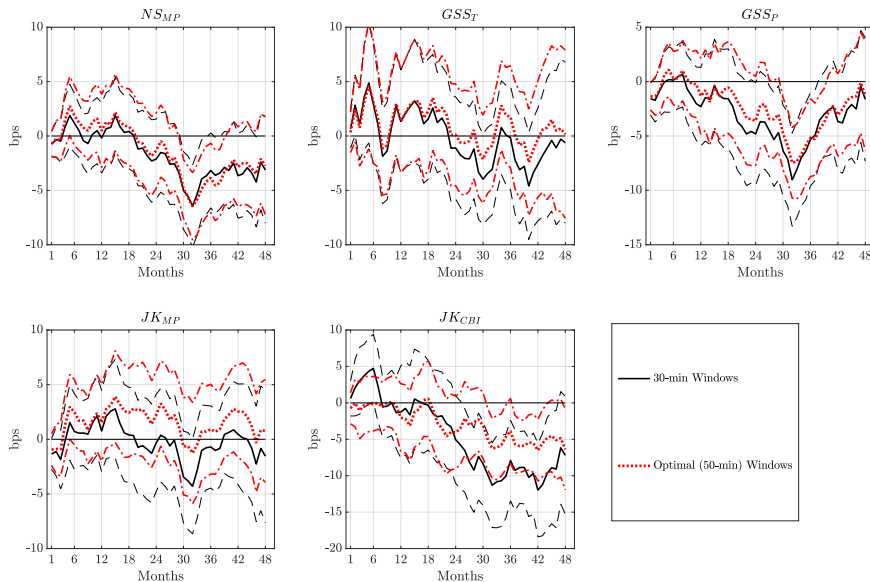
MP Shocks: Impulse Responses for EBP

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MP Shocks: Impulse Responses for 2Y Treasury Yield

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“Joint” and “One Signal” Approaches

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- ▶ **Recap:** XLNet-Base approx $f(\text{FOMC Statement}) = DP_{t+n}$, $\forall$ Folds of $\forall DP_{t+n}$
→ “Joint” estimation of signal and optimal event window length

“Joint” and “One Signal” Approaches

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“Joint” and “One Signal” Approaches

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→ “Joint” estimation of signal and optimal event window length
- ▶ Fine-tuning XLNet-Base for “joint” estimation = Computationally intensive
 - GPU + Financial constraints = Estimate optimal window lengths only up to $t + 60$
 - Current computation time: 249+ days

“Joint” and “One Signal” Approaches

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- ▶ **Recap:** XLNet-Base approx $f(\text{FOMC Statement}) = DP_{t+n}$, $\forall$ Folds of $\forall DP_{t+n}$
 - “Joint” estimation of signal and optimal event window length
- ▶ Fine-tuning XLNet-Base for “joint” estimation = Computationally intensive
 - GPU + Financial constraints = Estimate optimal window lengths only up to $t + 60$
 - Current computation time: 249+ days
- ▶ **Assumption:** NN Predictions in “joint-estimated” event window = Constant $\forall t$ ^{||}
 1. Much less computationally intensive
 2. Can check if FOMC statement characteristics affect optimal window length
 3. Can check if $\exists$ greater out-of-sample R_{OOS}^2 for $t + n > t + 60$

[Robustness Check](#)

^{||}Signal from XLNet-Base is likely to change $\therefore$ Changing LHS $DP_{t+n} \rightarrow$ retraining NN + “Joint” estimation was performed on “general” sample of FOMC statements, not specific types of statements.

Effects of FOMC Statement Characteristics on Event Windows

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► Use “one signal” approach to compare MSEs computed based on:

1. Complexity of FOMC statements
2. Similarity of FOMC statements
3. Presence of Dissents

[Complexity](#)[Similarity](#)[Dissents](#)

FOMC Statement Characteristics: Text Complexity (1/3)

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- ▶ Condition FOMC statements based on **semantic** complexity
 - Hernandez-Murillo and Shell (2014); Hansen and McMahon (2016); Smales and Apergis (2017); Haldane and McMahon (2018); De Pooter (2021); McMahon (2023); and others...

FOMC Statement Characteristics: Text Complexity (1/3)

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- ▶ Measured based on **Flesch Kincaid Grade Level** [Summary Stats](#)
 - Based on sentence structure, word structure, and word phonology

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 - Based on sentence structure, word structure, and word phonology
 - Range of reading Levels: 12.2–21.3
 - Median Reading Level: 16.5

FOMC Statement Characteristics: Text Complexity (1/3)

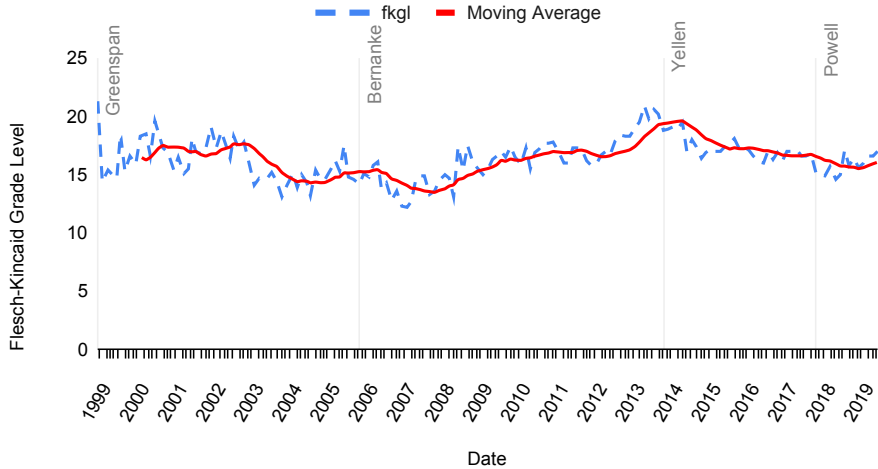
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- ▶ Measured based on **Flesch Kincaid Grade Level** [Summary Stats](#)
 - Based on sentence structure, word structure, and word phonology
 - Range of reading Levels: 12.2–21.3
 - Median Reading Level: 16.5
- ▶ Split sample conditioned on being $\leq$ or $>$ 16.5
- ▶ Calculate sub-set MSEs and event window lengths

FOMC Statement Characteristics: Text Complexity (2/3)

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Flesch-Kincaid Grade Level Readability of FOMC Statements



FOMC Statement Characteristics: Text Complexity (3/3)

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Metric	Simple	Complicated
<i>Minimised MSE</i>		
Average	1.25e-5	1.06e-5
<i>Event Window Length (Minutes)</i>		
Average	60	71

Table 14: Average MSEs and Event Window Lengths Using the “One Signal” Approach, Conditioned by FOMC Statement Complexity

→ FOMC statements with ↑ complexity → Longer event window on average

FOMC Statement Characteristics: Contextual Similarity (1/3)

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- ▶ Condition FOMC statements based on **contextual** similarity
 - Acosta and Meade (2015); Handlan (2022a); and others...

FOMC Statement Characteristics: Contextual Similarity (1/3)

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- ▶ Condition FOMC statements based on **contextual** similarity
 - Acosta and Meade (2015); Handlan (2022a); and others...
- ▶ Measured using a pre-trained, non-fine-tuned XLNet-Base model
 - Map complete statements into 768-dimensional dense vectors
 1. Processes text bi-directionally to capture context and syntax
 2. Ensures semantic signals are **out-of-sample** and **orthogonal** to asset price reactions

FOMC Statement Characteristics: Contextual Similarity (1/3)

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- ▶ Condition FOMC statements based on **contextual** similarity
 - Acosta and Meade (2015); Handlan (2022a); and others...
- ▶ Measured using a pre-trained, non-fine-tuned XLNet-Base model
 - Map complete statements into 768-dimensional dense vectors
 1. Processes text bi-directionally to capture context and syntax
 2. Ensures semantic signals are **out-of-sample** and **orthogonal** to asset price reactions
- ▶ $\Rightarrow$ Degree of similarity between 2 FOMC statements = **Cosine Similarity**:

Similarity Matrix

$$S^{A,B} := \cos \theta = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\| \|\mathbf{B}\|},$$

- ▶ **Scale**: Entirely different = $0 \leq \text{Cosine Similarity} \leq 1$ = Exact same

FOMC Statement Characteristic: Contextual Similarity (2/3)

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- ▶ $S^1 := (d, d - 1)$: Degree of similarity between **sequential** FOMC statements

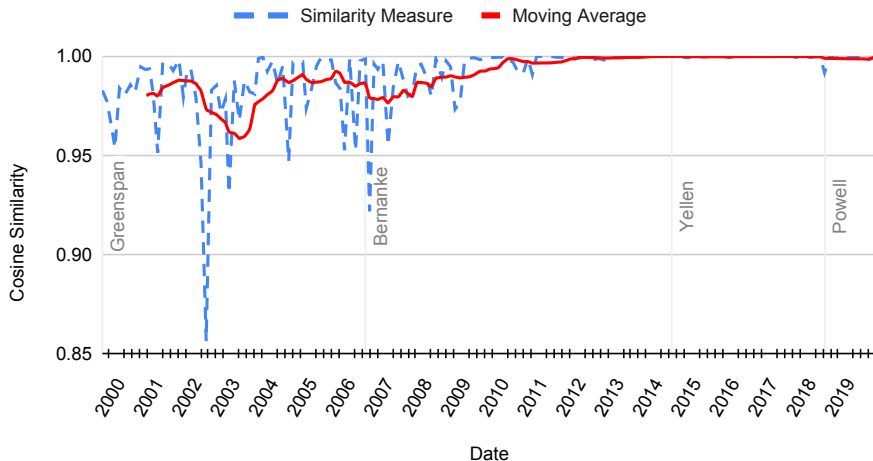
Summary Stats

- Range of S^1 : 0.856–0.999
 - 33rd percentile of S^1 : 0.994
- ▶ Split sample conditioned on being $\leq$ or $>$ 0.994
 - ▶ Calculate sub-set MSEs and event window lengths

FOMC Statement Characteristic: Contextual Similarity (2/3)

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Cosine Similarity of Sequential FOMC Statements



FOMC Statement Characteristics: Contextual Similarity (3/3)

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Metric	Different	Similar
<i>Minimised MSE</i>		
Average	9.99e-6	9.66e-6
<i>Event Window Length (Minutes)</i>		
Average	81	52

Table 15: Average MSEs and Event Window Lengths Using the “One Signal” Approach, Conditioned by FOMC Statement Similarity

→ Less similar FOMC statements → Longer event windows on average

FOMC Statement Characteristics: Presence of Dissents (1/2)

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- ▶ Condition FOMC statements based presence of dissent votes or not**
 - Riboni and Ruge-Murcia (2014); Tsang and Yang (2024); Bobrov et al. (2025); and others...

**I do not record and distinguish between dissent votes for tighter policy, easier, policy, or other reasons.

FOMC Statement Characteristics: Presence of Dissents (1/2)

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- ▶ Condition FOMC statements based presence of dissent votes or not**
 - Riboni and Ruge-Murcia (2014); Tsang and Yang (2024); Bobrov et al. (2025); and others...
- ▶ Roughly 40% of FOMC statement sample has recorded dissents
- ▶ By Fed tradition, dissents usually recorded if majority opinion = unacceptable
- ▶ Presence of dissents provides additional info. for markets to process

**I do not record and distinguish between dissent votes for tighter policy, easier, policy, or other reasons.

FOMC Statement Characteristics: Presence of Dissents (1/2)

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**I do not record and distinguish between dissent votes for tighter policy, easier, policy, or other reasons.

FOMC Statement Characteristics: Presence of Dissents (2/2)

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Metric	Unity	Dissents
<i>Minimised MSE</i>		
Average	9.57e-6	1.43e-5
<i>Event Window Length (Minutes)</i>		
Average	53	83

Table 16: Average MSEs and Event Window Lengths Using the “One Signal” Approach, Conditioned by Presence of Dissents in FOMC Statements

→ FOMC statements with dissents → longer event windows on average

FOMC Statement Characteristics: Summary Table

Complexity S1 Similarity

Metric	FKGL	S^1
Count	165	164
Mean	16.361	0.991
SD	1.715	0.017
Max	21.3	0.999
75 th	17.3	0.999
Median	16.5	0.998
25 th	15.1	0.987
Min	12.2	0.856

Table 17: Descriptive Statistics for Heterogeneity Analyses

Robustness Check of Optimal Event Windows

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1. Pick an interest-rate or equity futures contract

^{††}Performed for *FF2*, *FF4*, *TUc1*, *TYc2*, *USc1*.

Robustness Check of Optimal Event Windows

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1. Pick an interest-rate or equity futures contract
2. Take predictions $\widehat{DP_{t+n}}$ for each $k = 5$ fold from optimal event window

^{††}Performed for *FF2*, *FF4*, *TUc1*, *TYc2*, *USc1*.

Robustness Check of Optimal Event Windows

[Back to One Signal](#)

1. Pick an interest-rate or equity futures contract
2. Take predictions $\widehat{DP_{t+n}}$ for each $k = 5$ fold from optimal event window
3. Check if $\overline{R_{OOS}^2} \forall t + n \geq \overline{R_{OOS}^2}$ in optimal window length
 - Yes: Perform “joint” estimation in that window length^{††}
 - No: Go back to step 1

^{††}Performed for *FF2*, *FF4*, *TUc1*, *TYc2*, *USc1*.

Robustness Check of Optimal Event Windows

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 - Yes: Perform “joint” estimation in that window length^{††}
 - No: Go back to step 1

► Results:

1. Optimal window length has highest $\overline{R_{OOS}^2}$
 2. “Jointly” estimated $\overline{R_{OOS}^2}$ for window $> t + 60$ greater than “” for window $t + 20$
- Event window with global maximum $\overline{R_{OOS}^2}$ **could be** in window length $> t + 60$

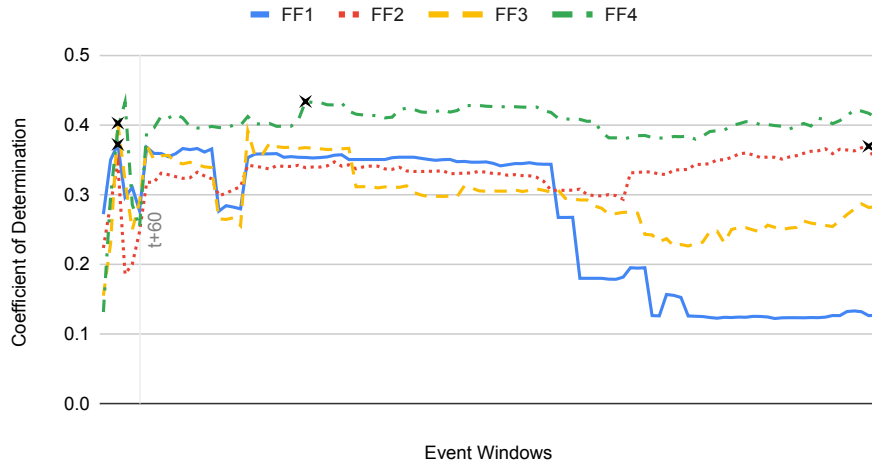
^{††}Performed for *FF2, FF4, TUC1, TYC2, USC1*.

Testing R^2 Using “One Signal” Approach for Federal Funds Futures

FF2

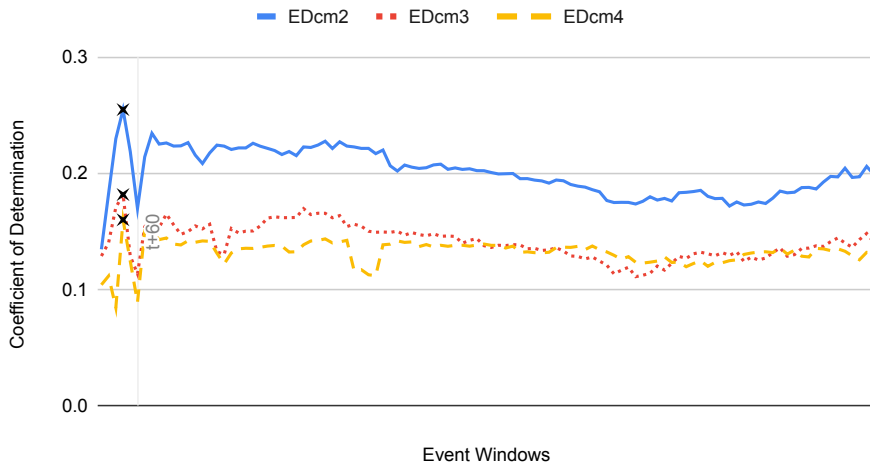
FF4

Out-of-sample R^2 Using "One Signal" Approach (FFFs)



Testing R^2 Using “One Signal” Approach for Eurodollar Futures

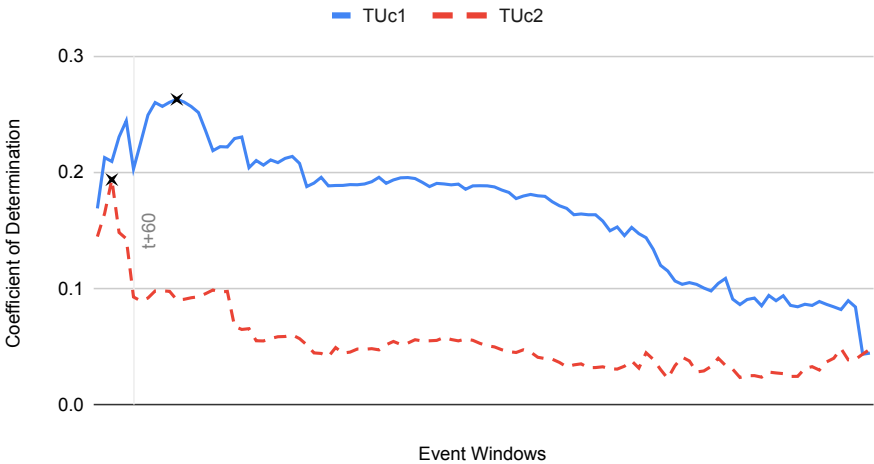
Out-of-sample R^2 Using "One Signal" Approach (EDs)



Testing R^2 Using “One Signal” Approach for 2-Year Treasury Futures

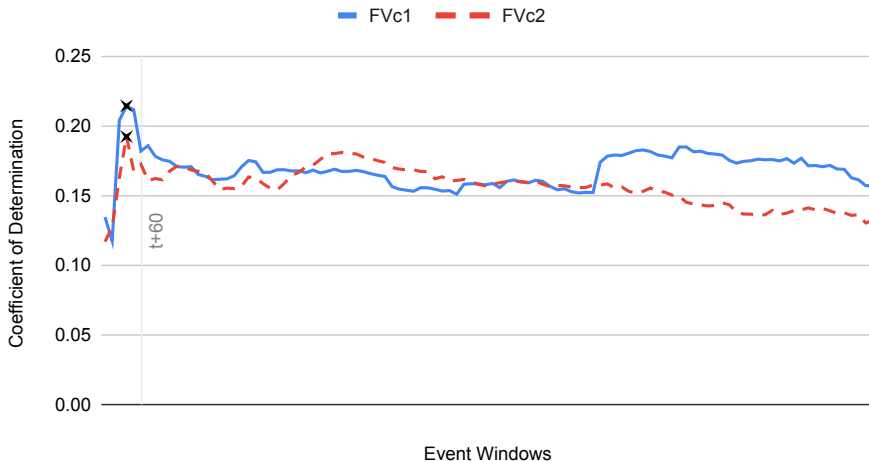
TUc1

Out-of-sample R^2 Using "One Signal" Approach (TUs)



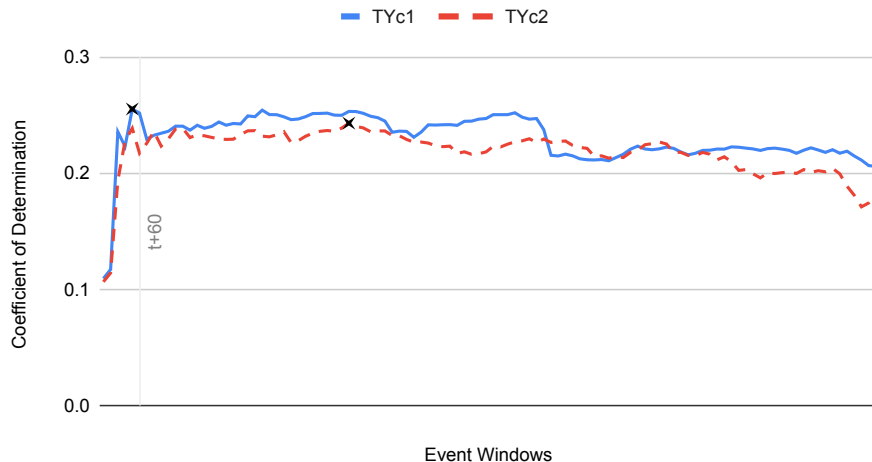
Testing R^2 Using “One Signal” Approach for 5-Year Treasury Futures

Out-of-sample R^2 Using "One Signal" Approach (FVs)



Testing R^2 Using “One Signal” Approach for 10-Year Treasury Futures TYc2

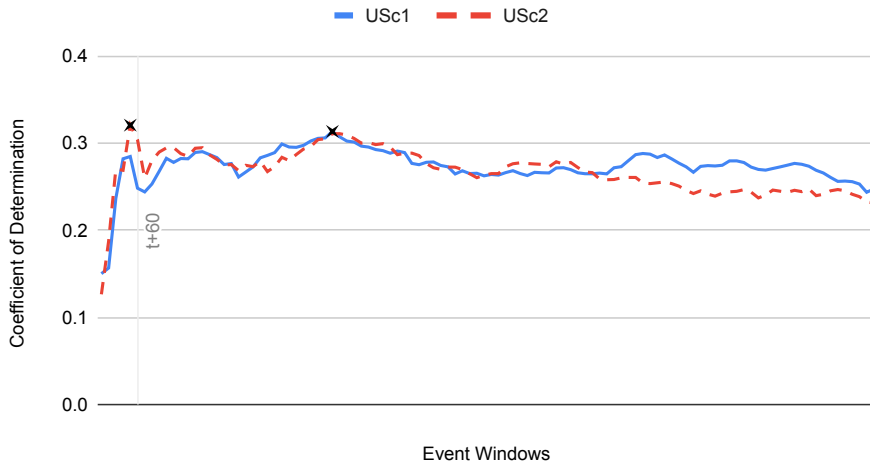
Out-of-sample R^2 Using "One Signal" Approach (TYs)



Testing R^2 Using “One Signal” Approach for 30-Year Treasury Futures

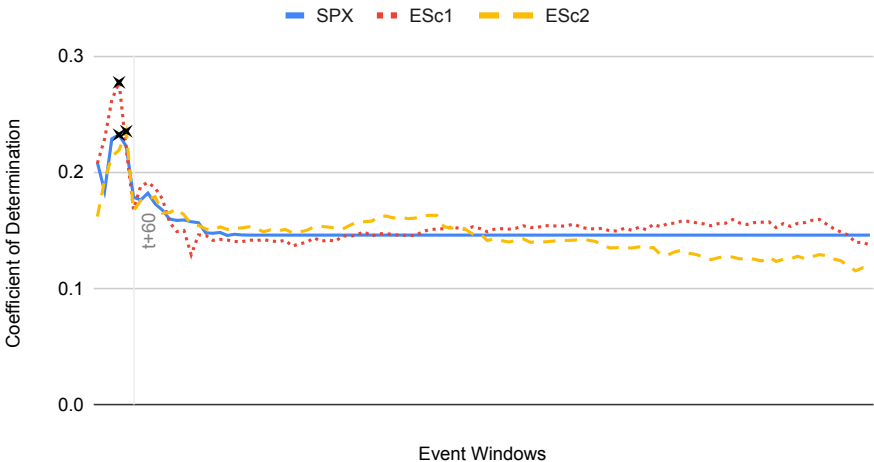
USc1

Out-of-sample R^2 Using "One Signal" Approach (USs)



Testing R^2 Using “One Signal” Approach for S&P Index

Out-of-sample R^2 Using "One Signal" Approach (S&P 500)



Cosine Similarity Matrix

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